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Nail-Mounted Haptic Cues for Piano Training and Tele-training

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Abstract

Skilled hand movement depends on choosing the correct finger as well as the correct target, yet guidance technology poorly supports that choice. Screens and sounds show which key to press and when, but neither directly specifies the acting digit. MIDI likewise records the key and timing, not the finger used, so adherence to the prescribed fingering cannot be measured from MIDI alone. This work addresses both gaps. A platform delivered vibration to the nail of the target finger while leaving the fingertip free to sense the key, with per-key illumination and camera-based finger identification aligned to MIDI. Bench tests selected the actuator, mounting, and drive settings before recruitment. A constraint-driven generator then produced sequences of measured, ordered difficulty matched across conditions. Twenty participants performed the task with key-only, visual-finger, and vibrotactile-finger guidance. Relative to matched visual cues, vibrotactile cues reduced reaction time by 247 ms and improved Complete-action Accuracy by 3.65 percentage points. Benefits varied across digits, and wrong-hand errors fell from 28.8% to 2.7% of substitutions, while key accuracy remained near ceiling in both conditions. The digit and error patterns are consistent with reduced finger-selection demands, although selection and execution were not measured separately. The common representation of keys and fingers also drove a working remote-teaching deployment supporting live and recorded guidance. The evidence is confined to immediate cue-guided performance within one session; retention after cue withdrawal and behavioural benefit during remote use remain untested.

Acknowledgements

I would like to thank Dr Alexis W.M. Devillard and Dr Etienne Burdet for their guidance throughout the project. Their advice not only established the technical direction of the work but also guided the development of the experimental platform; furthermore, the feedback they provided during project implementation and report writing helped to further refine both aspects of the work.

List of Abbreviations

Abbreviation	Meaning
AC	Alternating current
ANOVA	Analysis of variance
CA	Complete-action Accuracy
CI	Confidence interval
CS	Chip select
CUSUM	Cumulative sum
DC	Direct current
DMA	Direct memory access
ERM	Eccentric rotating mass
FA _{cond}	Conditional Finger Accuracy (among correct-key responses)
GRB	Green–red–blue (colour-channel order)
HTTPS	Hypertext Transfer Protocol Secure
IQR	Interquartile range
LED	Light-emitting diode
LOPO	Leave-one-participant-out
LRA	Linear resonant actuator
MIDI	Musical Instrument Digital Interface
MISO	Master in, slave out
MOSI	Master out, slave in
PCB	Printed circuit board
p95	95th percentile
PWM	Pulse-width modulation
RQ	Research question
RMS	Root mean square
RT	Reaction time
RTT	Round-trip time
SCK	Serial clock
SD	Standard deviation
SPI	Serial Peripheral Interface
THD	Total harmonic distortion
TLS	Transport Layer Security
UART	Universal Asynchronous Receiver–Transmitter
USB	Universal Serial Bus
UVC	USB Video Class

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Chapter 1

Introduction

1.1 Motivation and Related Work

Keyboard performance requires identifying a target key, assigning a finger, and executing the movement. Fingering determines whether the next note remains accessible, so poor choices can propagate through a sequence. The digits are neither mechanically nor neurally independent; passive coupling particularly limits the index, middle, and ring fingers [1, 2]. Selecting a specified digit also recurs in fine-motor training and rehabilitation, where keyboard practice provides structured, quantifiable exercise [3, 4, 5]. Here the keyboard serves as an instrumented model of finger guidance, discretising responses, timestamping events, and making finger selection scorable.

Augmented feedback can improve immediate performance without producing lasting learning, and its channel determines what it can express [6, 7]. Visual displays suit spatial targets and auditory feedback conveys timing or outcome; neither directly specifies the acting digit, requiring fingering inference or symbolic decoding. These mappings operate within a sensorimotor loop governed by predicted and actual sensory consequences [8, 9]. A bodily haptic cue can specify the effector through its location. Vibrotactile stimulation reduced novice pianists' timing error relative to visual stimulation, although note-accuracy evidence was unresolved [10]. Comparisons with electrical stimulation and linkage devices reveal modality-specific trade-offs [11], while explicit error feedback can overcome ceilings that repetition alone does not [12]. Whether finger-addressed cues outperform matched visual finger information remains open.

Digital piano systems support home rehabilitation, web-based teaching, and mobile learning [5, 13, 14], yet typically indicate what to press without specifying which finger should act. Remotely, teachers cannot physically reposition the student's hand and must communicate finger identity through available channels. This work supplies that cue locally and extends it to tele-training.

Two limitations remain. The first is cue placement. Most wearable haptics signal timing, error, or a global hand/body prompt rather than digit identity [15, 11]. Finger-specific actuation must avoid disrupting the trained action. Although the fingertip has the hand's highest tactile innervation density [16], an actuator there obstructs the surface sensing key edge, contact, and release [17]. Proximal-phalanx, dorsal, and passive-learning designs therefore move actuation off the pad [18, 19, 20, 21].

The fingernail offers a rigid, repeatable mounting surface near the pad but outside its contact patch. Nail sensibility and localisation [22, 23], together with nail-worn and fingerpad-free devices [24, 25, 17], support this form factor. Referral towards the fingertip could place perception near the action while leaving contact free, but neither this study nor the cited literature establishes referral during active key contact. It remains a design rationale.

The second limitation is measurement. MIDI records keys and timing, not fingers. Nail-worn and fingerpad-free devices establish mounting feasibility [24, 25, 17], while the piano comparison supplied finger identity through vibration but only key identity visually [10]. The remaining gap is to evaluate ten-finger tactile guidance that preserves fingertip contact against equally informative visual cues, measuring the acting finger rather than inferring it from MIDI.

1.2 Aim, Research Questions, and Objectives

This project addresses that combined design and measurement gap. It aims to build nail-mounted finger guidance that preserves natural key contact, evaluate it against matched visual finger cues under controlled bimanual sequence difficulty, and extend the same representation of keys and fingers to tele-training.

- **RQ1.** With target-finger information and difficulty matched, do vibrotactile and visual cues differ in reaction time and Complete-action Accuracy?
- **RQ2.** Do effects vary across homologous fingers 1–5, and do the modalities produce different finger errors?

The objectives are to **(O1)** integrate MIDI, illuminated keys, nail actuators, camera recognition, and synchronised logs; **(O2)** validate actuator, attachment, and drive choices; **(O3)** generate seeded sequences with ordered difficulty matched across conditions; **(O4)** compare key-only, visual-finger, and vibrotactile-finger guidance within participants, overall and by digit; and **(O5)** implement live and recorded tele-training by reproducing a teacher’s recognised actions through the student’s cues.

Generated sequences prevent repertoire familiarity and auditory memory from substituting for the cue [26, 27]. Musical expression and key-mechanics rendering are outside scope. Evidence concerns immediate guided performance, not retention after cue withdrawal. Tele-training is a platform deployment, not an experimental condition: it received no user study, so remote behavioural benefits remain untested.

Chapter 2

Methods

2.1 System Implementation

The platform (Figure 2.1) used a Nektar SE25 MIDI keyboard with 25 calibrated keys (notes 48–72). Two under-key WS2812 strips cued the target key without altering key feel; ten nail-mounted actuators cued the fingers. A Teensy 4.1 and two PWM boards drove the actuators from a motor supply isolated from logic power.

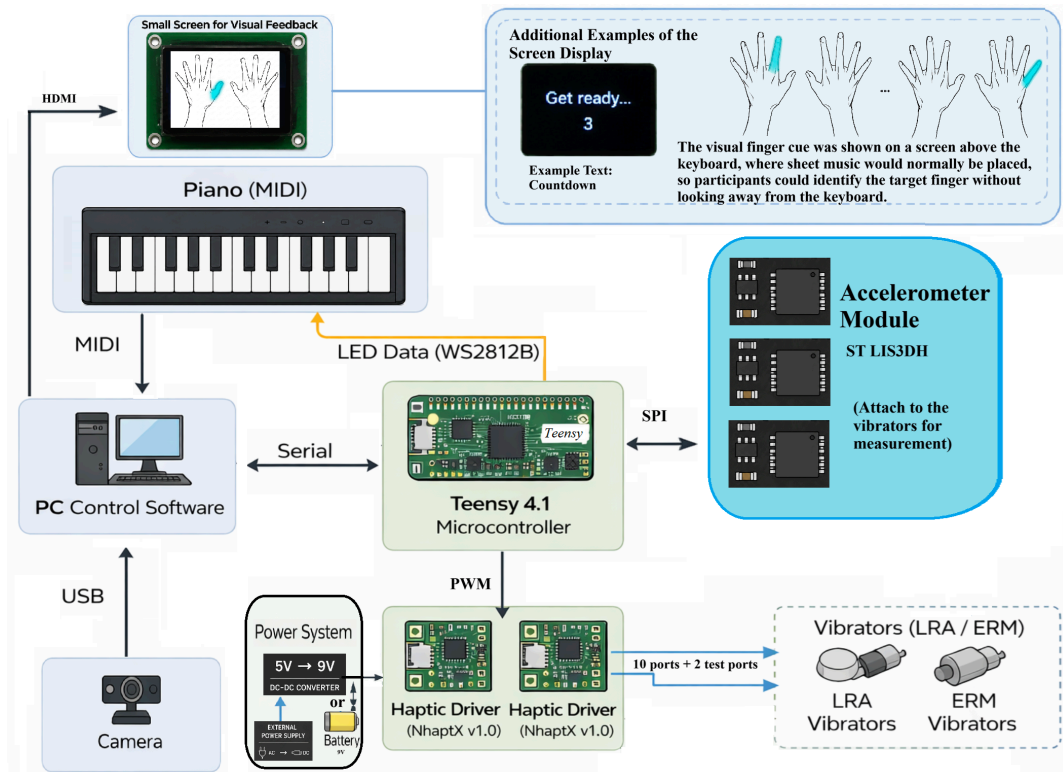


Figure 2.1. Ten-finger hardware architecture. MIDI and video return separately from Teensy cue delivery. Twelve motor channels include two spare outputs. The 9 V rail was supplied by a battery or, here, a converter from 5 V. Accelerometers were used only for bench validation and were absent from participant sessions.

Python software scheduled trials, drove cues, logged MIDI, and recorded 1280×720 , 30-fps video. All streams received host-clock Unix timestamps. Appendix Table C.1 records the configuration; Appendix Figure C.1 shows the 3D-printed enclosure, integrated electronics, and bench and session interfaces.

2.2 Hardware Validation and Final Configuration

Three pre-recruitment bench experiments fixed the configuration. A coupled accelerometer measured demeaned three-axis vector RMS. Frequency was set before amplitude, giving a fixed 224-Hz, 25.1%-duty PWM operating point. The selected duty was the measured point closest to a provisional target band rescaled from an earlier project metric; it defined a reproducible setting, not a psychophysically calibrated intensity [28]. LRA and ERM onset latencies were compared over ten trials each using CUSUM detection [29]. Double-sided tape, Blu Tack, and eyelash adhesive were compared over three independent five-trial mounts, with mount as the averaging unit. Only transmitted intensity was measured; handling was judged rather than scored. The rig, sweeps, selection calculation, detection settings, and attachment metrics are in Appendix B.

2.3 Main User Study

2.3.1 Integrated Operation and Synchronisation

Each event illuminated the target key and, where applicable, displayed or vibrated the target finger. The visual finger cue appeared on a small display immediately above the keyboard, in the usual sheet-music position. Cue, keys, and hands remained within one field of view, allowing participants to read the cue without looking away from the keys. The dispatch callback timestamped cue onset. The first MIDI note-on ended the event; incorrect presses counted as errors without retries. Events timed out after 5 s, with 400 ms between events.

Camera start-up latency prevented direct conversion from host time to frame number: the first-frame offset varied by roughly -0.2 to $+0.55$ s, or up to sixteen frames. A keyboard LED flash after recording began supplied a host timestamp and video frame to re-anchor each trial. Pulse-duration and edge-agreement checks rejected false anchors (Appendix B).

2.3.2 Controlled Stimulus Sequence Generator

Stimuli followed a constrained serial-response design to avoid repertoire familiarity and auditory or rhythmic confounds [26, 27, 30, 31]. Each sequence contained 30 single actions specifying a hand, finger, and key, with both hands represented. Fifteen white keys were used because black keys change reach and posture [32].

Difficulty combined motor cost C_m , sequence complexity C_s , and bimanual coordination C_c . Level α used short, predictable movements within separate hand regions. Level β increased movement and transition uncertainty and allowed both hands to reach the midpoint, without crossing. Level γ required longer, more varied movements and more frequent hand alternation, with bounded crossing. Appendix Table B.2 gives the numerical constraints.

Seeded rejection sampling and family matching produced comparable sequences within each level. Before locking, component medians had to follow the intended ordering with fewer than 10% adjacent-level violations. Choice-reaction accounts [33] and Fitts' law [32] motivated transition uncertainty and displacement. Definitions, rejection rules, and tolerances are in Appendix B.

2.3.3 Camera-Based Finger Identification

Because MIDI identifies the key and time but not the acting finger, calibrated key regions and the two-hand MediaPipe landmark model [34] supplied finger evidence without markers, gloves, or depth sensors. Edge-

constrained calibration followed each visible key shape instead of assuming rectangles, and assignment used the pressed key rather than the cued target. For visible fingertips $\mathcal{V}_t$, pixel distances $d_{f,t}$ to that key yielded a softmax distribution [35, 36]:

$$p_t(f) = \frac{\exp(-d_{f,t}/\tau)}{\sum_{f' \in \mathcal{V}_t} \exp(-d_{f',t}/\tau)}, \quad F_t = I(p_t(f_t) \geq \theta).$$

The target finger was scored correct when its mass reached θ . This graded rule avoided treating a near-tie as participant error merely because another fingertip was marginally nearer. Figure 2.2 shows one review frame. Before data collection, $\tau = 25$ px and $\theta = 0.40$ were fixed against key width and near-tie behaviour (Appendix B).

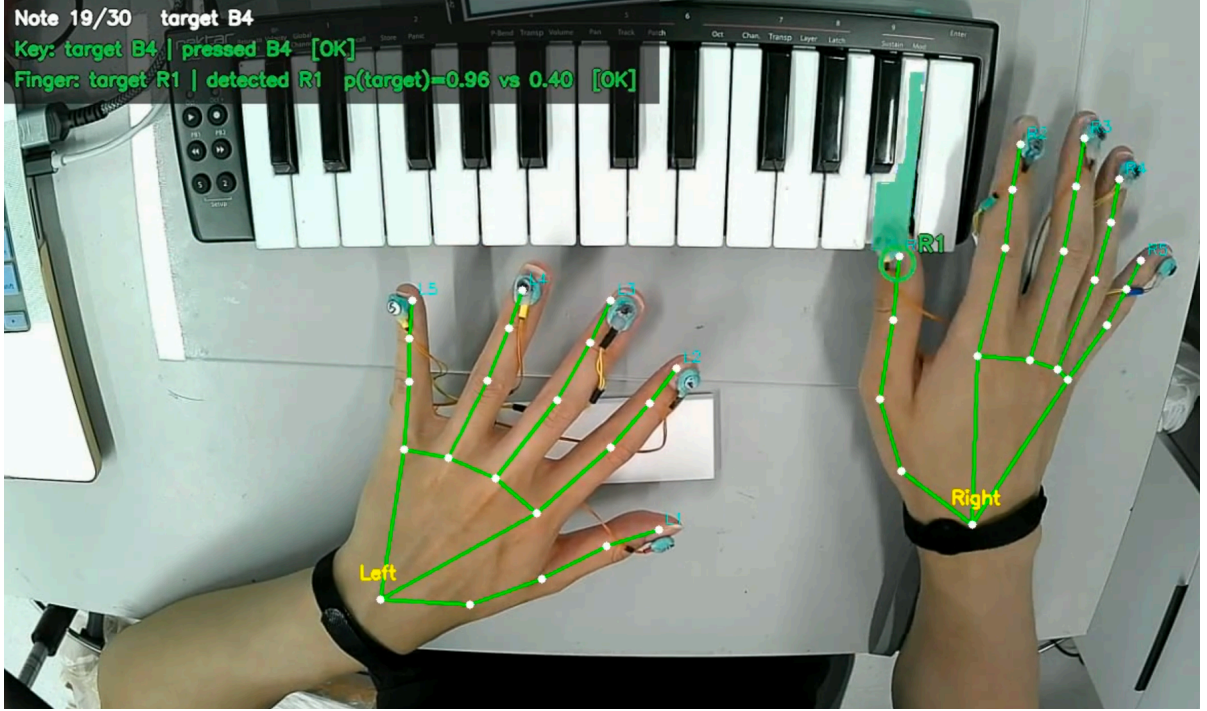


Figure 2.2. One review frame showing labelled MediaPipe fingertips, the pressed key region, and target-finger mass relative to $\theta = 0.40$.

Automated assignment was a pre-screen. One reviewer inspected every event frame by frame, corrected misidentified finger labels, and left occluded events unresolved. Group analysis then used these reviewed labels with the defined scoring rules. Every scored event had visual evidence; no second rater cross-checked the labels.

2.3.4 Design and Procedure

Twenty participants completed the study (19 male, one female; mean age 22.9 years, SD 2.2, range 20–28; 16 right- and four left-handed). Sixteen reported no keyboard experience and four reported 1–10 years; experience and handedness were metadata, not eligibility criteria or analysis factors.

The within-participant design used three distinct sequences per condition and difficulty level (Figure 2.3), giving 27 trials and 810 target events. Every condition backlit the target key. Participants wore the actuators throughout, so only finger-cue availability and modality changed. LED output was fixed at no less than 3:1 contrast [37]. Condition C vibrated continuously until keypress or timeout, matching the visual cue’s envelope rather than adding pulse rate as a perceptual variable [38]. Keyboard sound and all other audio output were muted throughout the main study to remove auditory-motor feedback and acoustic interference [26, 27].

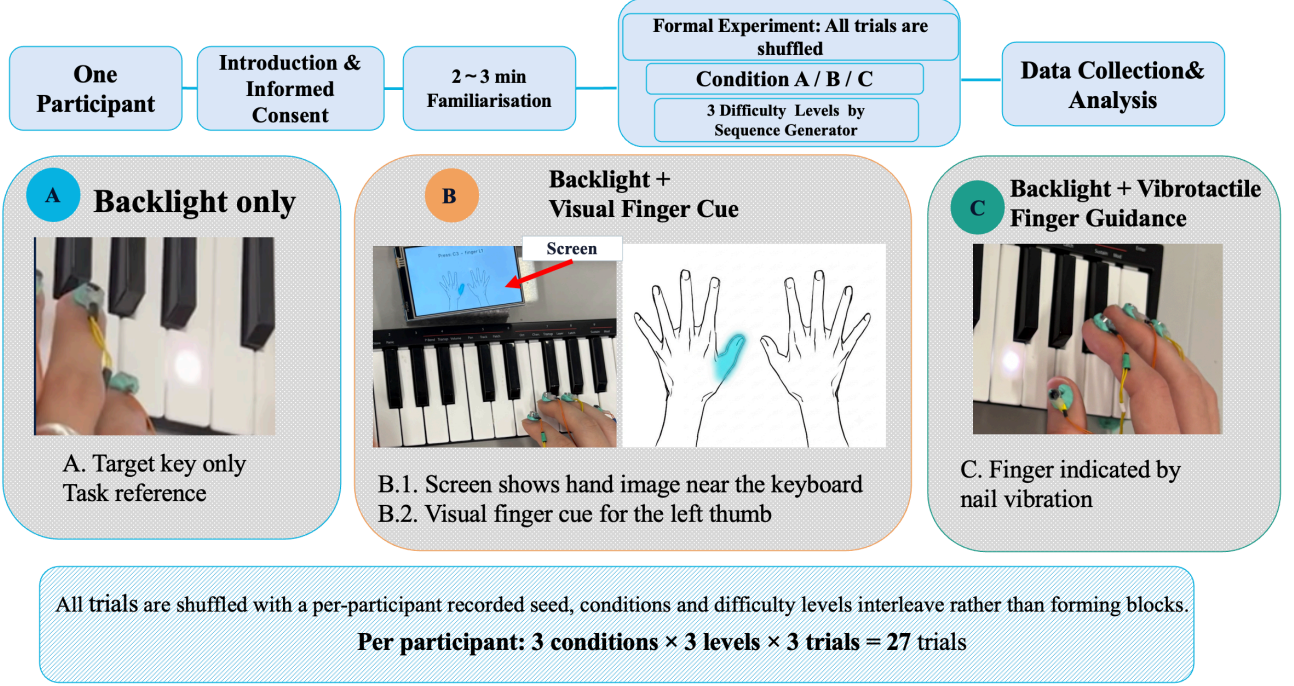


Figure 2.3. Study design and procedure. Top shows session flow; middle defines Conditions A–C; bottom shows the $3 \times 3 \times 3$ trial structure for each of 20 participants. Actuator fitting and the ten-finger check preceded familiarisation.

Both B and C specified the target finger, differing only in visual versus nail-vibration delivery. Condition A provided no finger cue; agreement with its hidden target served as an unguided-choice reference, not a modality baseline.

For each participant, a recorded seed was used to randomly allocate each level’s nine sequences without replacement to A, B, and C (three each), then shuffle all 27 trials, interleaving conditions and difficulty levels. A randomised ten-finger check had to be completed without error before testing. Bench height was adjusted per participant [39, 40]. Between trials, participants could request a self-paced break of up to 5 min [41, 42].

2.3.5 Outcome Measures and Analysis

For event t , MIDI gave key correctness $K_t = I(\hat{m}_t = m_t)$, the threshold gave finger correctness F_t , and $A_t = I(K_t \wedge F_t)$ denoted a complete action. Primary outcomes were Complete-action Accuracy, $CA = \sum_t A_t / 30$, and cue-to-keypress reaction time for $A_t = 1$ events; every ANOVA used this reaction time. Auxiliary conditional Finger Accuracy ($FA_{\text{cond}} = \sum_t A_t / \sum_t K_t$) isolated finger selection among correct-key responses, while correct-key reaction time, error composition, timeouts, and no-majority fingertip events checked robustness and data quality.

Measures were aggregated within participant by condition, difficulty, sequence, and repetition. Participants remained the unit of inference. Paired comparisons of B and C used two-sided t -tests, 95% participant-level percentile-bootstrap CIs, Cohen’s d_z , and Wilcoxon signed-rank checks. Group-mean intervals used the same bootstrap, keeping accuracy bounds within 0 and 100%.

The pre-specified reaction-time ANOVA crossed Condition (B, C) with Finger ID (1–5). Greenhouse–Geisser correction was applied to Finger and the interaction to address possible departures from sphericity. Secondary

models replaced Finger with difficulty or repetition, using the same complete-action reaction time and complete-case participant grid without multiplicity correction. Bounded, ceiling-compressed accuracy was excluded from ANOVA. Participant contrasts, per-finger cells, and automatic-score sensitivity across $\theta \in \{0.30, \dots, 0.50\}$ were reported.

Binomial sensitivity models used cell successes over judged events, a logit link, and a participant random intercept. An additive Condition + Finger model estimated the overall odds ratio; a Condition $\times$ Finger model estimated digit-specific effects. Maximum likelihood used adaptive Gauss–Hermite quadrature. Likelihood-ratio tests assessed model terms; odds ratios used participant-clustered robust intervals with a $t(N - 1)$ reference. The participant-level paired contrast remained primary. Figure stars denote $^*p < 0.05$ and $^{**}p < 0.01$ for the captioned test. Unmarked effects do not meet 0.05, or the panel is descriptive.

An exploratory ten-finger model combined habitual, motor, and cue evidence using leave-one-participant-out folds and a multinomial-logistic benchmark (Appendix C.2).

2.4 Tele-training Deployment

The representation of keys and fingers was extended to teacher and student clients through a relay server (Figure 2.4). Both stations were self-contained, with separate keyboards, cameras, and cue hardware; their computers exchanged events only through the network. In synchronous mode, the student’s key light, screen, and actuators reproduced the teacher’s played key and camera-identified finger. Only the student wore nail actuators because the teacher generated rather than received cues. In asynchronous mode, a recorded plan was downloaded and scheduled locally.

Figure 3.1 shows the complete paired configuration, in which a divider and headphones limited visual and acoustic cross-talk. The main study used only the student station, with audio muted.

Reaction time began at the student’s local cue-ready instant, excluding network delay. Visual and haptic components were dispatched together locally.

This engineering demonstration, not a participant experiment, used 1,000 transport probes, alignment data from eleven two-machine sessions, and two single-operator modality demonstrations. Protocol and clock conventions are in Appendix B.

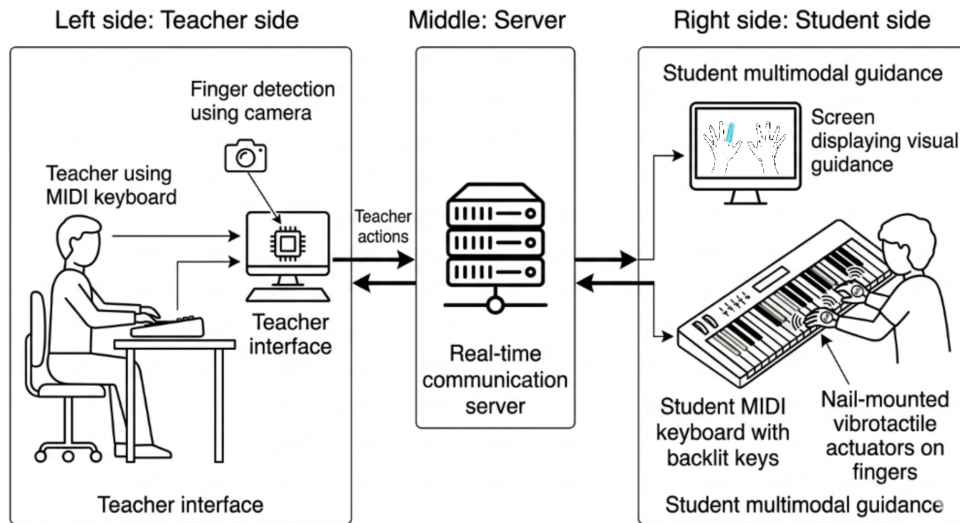


Figure 2.4. Two-station tele-training architecture. Events specifying keys and fingers crossed the relay, with cue timing and multimodal output generated locally.

2.5 Ethics, Consent, and Data Handling

2.5.1 Approval, Consent, and Participation

The work was conducted with approval from the Research Ethics Committee at Imperial College London for “Cross-modal platform to investigate multisensory integration during human machine interaction” (approval number 22IC7699), granted before recruitment. Participants received the purpose and procedure in advance and gave written consent for participation and data use. Participation was voluntary, with withdrawal permitted without penalty; there was no dependent relationship, coercive incentive, or deception.

2.5.2 Data Protection

Data are pseudonymised rather than anonymous: datasets and videos use participant codes, while the identity link is stored separately on secure institutional storage, consistent with the Data Protection Act 2018 [43]. Recordings exclude faces. Privacy and validity conflict because frame-level verification cannot be reconstructed from landmark streams alone; raw hand video is therefore retained only for the approved period under restricted access. Tele-training sends cue events over TLS-protected HTTPS and secure WebSocket, and the relay stores no video [44, 45].

2.5.3 Participant Safety

Standards for hand-transmitted vibration assume sustained power-tool contact at the grip. Their exposure margins were not derived for brief, high-frequency transmission through the nail [46, 47]; quoting them would give false assurance. The bounded protocol used fixed calibrated amplitude, event-limited cues, and monitoring for discomfort, warmth, irritation, or detachment to limit exposure and adaptation risk [48]. Adaptation was not measured. No adverse event occurred during the protocol, but this does not establish safety for longer use, which requires a dose study. Adhesive contacted only the nail, putty was discarded between participants, and

detachment was a stopping condition for both safety and cue validity.

2.5.4 Scientific Integrity and Institutional Responsibility

Automated finger labels were treated as a pre-screen. Every event was reviewed frame by frame and corrected where necessary. Seeded, locked stimuli, pre-fixed parameters, threshold sensitivity analysis, and reporting of ceiling compression, unmatched physical onset, and the small key-accuracy difference alongside favourable results protect against selective analysis. Code, materials, and parameter provenance support reproduction, challenge, and handover (Appendix B). No animal work was involved.

2.5.5 Stakeholders, Justice, and Wider Impact

Stakeholders include students, teachers, rehabilitation users [5, 49], researchers, and the institution. The small sample (20 people aged 20–28; 19 male) cannot support generalisation across age, sex, nail geometry, or hand morphology. Requiring nail-vibration perception and two-hand keyboard positioning may exclude people with neuropathy or impaired hand function, including intended rehabilitation users. The visual cue met 3:1 contrast [37]; components are commercially available, but affordability was not measured. Reusable electronics limit single-use waste, while disposable adhesive protects hygiene; no lifecycle or environmental assessment was performed.

Continuous hand video can become surveillance, so deployment should favour local processing or a non-video mode where possible. Future users should retain control over recordings and derived performance data rather than transferring ownership implicitly to a platform provider. Claims must also remain limited to immediate, single-session guided performance, not learning, retention, or clinical benefit. These choices enact the Engineering Council principles [50]: rigorous and auditable measurement, honesty about evidential limits, respect for law and public welfare, and reproducible documentation that permits criticism and extension.

Chapter 3

Results

3.1 Platform and Hardware Validation

Both stations were built and exchanged cue events over the relay (Figure 3.1). The main study used only the student station, completing 540 trials across 20 participants without protocol deviation while logging MIDI and video for offline scoring.

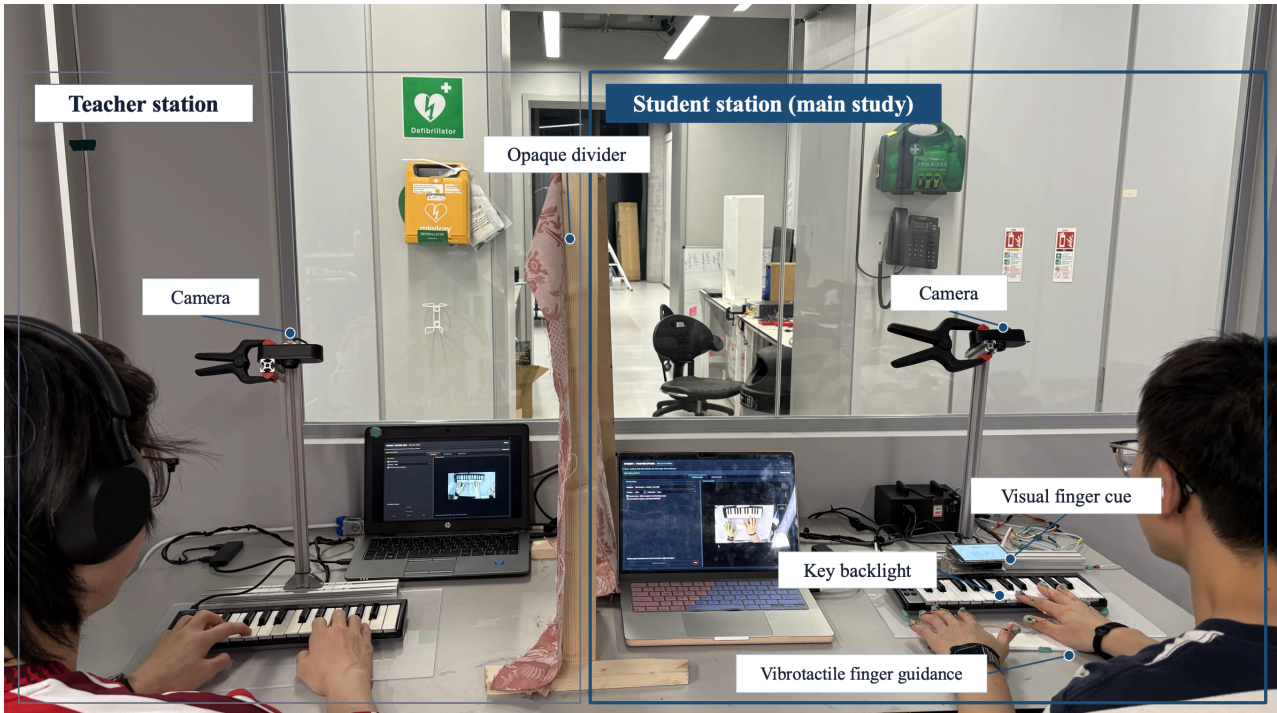


Figure 3.1. Two-station platform. The grey outline marks the teacher and the blue outline the student station, which provides visual finger cues, key lighting, and nail vibration. Both have overhead cameras, and the divider blocks direct sight. Only the student station was used in the main study.

3.1.1 Actuator Characterisation and Drive Calibration

The ERM output rose broadly with duty, whereas the LRA concentrated its output in a narrow frequency band, allowing intensity to vary at fixed frequency. A frequency sweep located the mounted resonance at 224 Hz, where the PWM-amplitude sweep selected the firmware parameter $\text{amp}_{\text{PWM}} = 64$ on its 0–255 scale. This parameter sets the duty cycle, $D = \text{amp}_{\text{PWM}}/255 = 0.251$ (25.1%; 1.12 ms high in each 4.46 ms period), rather than

specifying a sinusoidal voltage amplitude. Its measured vector-RMS acceleration, 2.68 m/s^2 , was closest to the provisional 2.7 m/s^2 target (Appendix Figure B.1c, d). The applied signal is a 0–9 V rectangular PWM waveform; the sine-shaped reference in panel (d) represents only its fundamental component at the drive frequency. These are mounted-system operating points, not perceptual constants [28].

3.1.2 Actuator Type

At those settings the two actuator types differed by more than an order of magnitude. Across ten trials each, motion onset was $6.14 \pm 1.20 \text{ ms}$ for the LRA against $205.93 \pm 22.11 \text{ ms}$ for the ERM, and command-to-steady intervals $127.6 \pm 11.6 \text{ ms}$ and $491.5 \pm 24.9 \text{ ms}$ (Figure 3.2). The ERM onset alone is roughly a quarter of the reaction times later observed and would enter measured latency as a condition-specific offset, so the LRA was adopted.

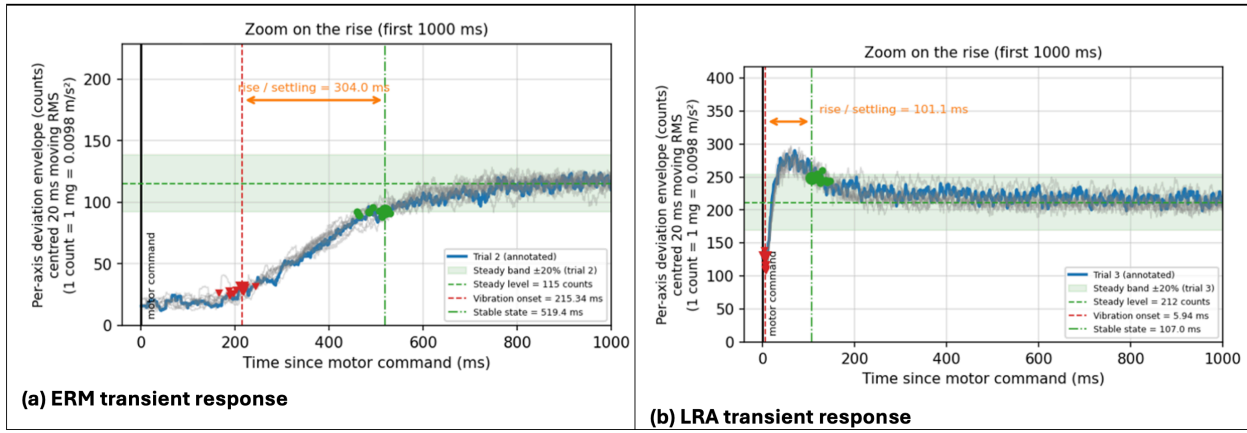


Figure 3.2. Actuator transient response, one representative trial of ten per actuator: (a) ERM and (b) LRA, each with the motor command, detected onset, steady level, and $\pm 20\%$ steady band. Actuator and accelerometer were rigidly coupled and suspended freely, so the amplitude (counts, or the m/s^2 equivalent at 1 count = 1 mg) is the free assembly’s own output rather than the vibration delivered through a nail mount.

3.1.3 Attachment Method

All three adhesives transmitted repeatable cues; double-sided tape transmitted most and cosmetic adhesive least, but transmitted intensity did not drive the choice. Blu Tack was selected because it holds immediately, repositions without residue, and can serve a whole fitting. Actuator, attachment, and drive were fixed before recruitment. The per-mount transfer measurements are reported in Appendix B (Table B.1; Figure B.2).

3.1.4 Stimulus Pool

The locked pool comprised nine sequences per difficulty level; all eleven components followed the intended ordering, adjacent-pair violations remained below the 10% limit, and every structural check passed (Appendix A).

3.2 Main User Study

The primary analysis compares matched visual (B) and vibrotactile (C) finger guidance; Condition A is retained only as an unguided descriptive reference.

3.2.1 Data Completeness

All 20 participants completed 540 trials and 16,200 recorded cue events. One carry-over was excluded, leaving 16,199 analysed events; manual review resolved every valid event. There were three timeouts in A, one in B, and none in C, leaving 16,195 scored keypresses, of which 4.50% had the automatic finger label overridden during that review (Appendix B, Figure B.3). Review used one rater without independent cross-checking. Most overrides involved near-ties, so this rate describes disagreement with the pre-screen, not participant error.

3.2.2 Condition Effects

Table 3.1 and Figure 3.3 show that C was faster and more accurate than B.

Table 3.1. Condition outcomes as mean $\pm$ SD across 20 participants. Primary outcomes are Complete-action Accuracy and complete-action reaction time. In A, finger-related entries measure agreement with hidden finger-ing, not accuracy.

Measure	A (key only)	B (visual finger)	C (vibrotactile finger)
Key Accuracy (%)	99.1 $\pm$ 0.8	98.9 $\pm$ 1.4	99.7 $\pm$ 0.4
Complete-action Accuracy (%)	18.8 $\pm$ 2.8	94.7 $\pm$ 3.1	98.3 $\pm$ 1.2
Finger Accuracy correct key (%)	19.0 $\pm$ 2.8	95.7 $\pm$ 2.5	98.6 $\pm$ 1.2
RT, correct-key events (ms)	488.2 $\pm$ 55.2	970.2 $\pm$ 124.1	723.5 $\pm$ 91.9
RT, complete-action events (ms)	488.8 $\pm$ 66.6	969.9 $\pm$ 123.4	723.2 $\pm$ 91.2

Participant trajectories show ceiling-level key accuracy and separation on finger-related outcomes (Appendix Figure D.1).

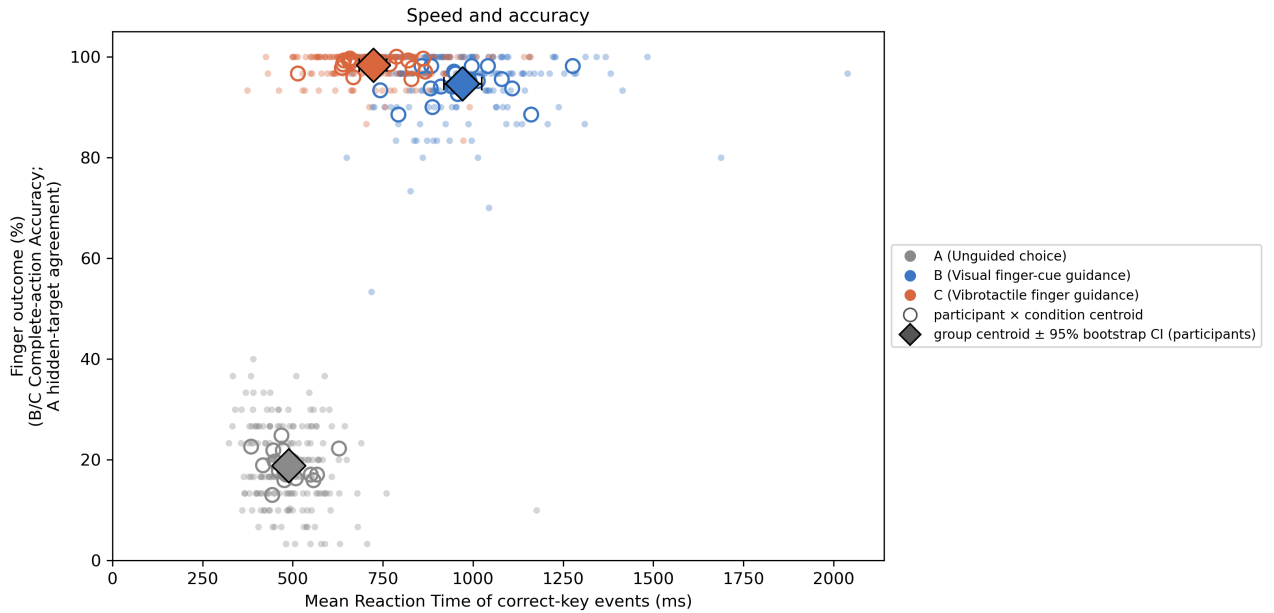


Figure 3.3. Speed and accuracy across 20 participants. Points are trials, rings participant-by-condition centroids, and diamonds group means with 95% participant-bootstrap CIs. The reaction-time axis starts at zero.

Complete-action Accuracy rose by 3.65 percentage points, 95% CI [2.39, 4.98], $t(19) = 5.35$, $p = 3.7 \times 10^{-5}$, $d_z = 1.20$, Wilcoxon $p = 0.00029$, with 19 of 20 participants higher under C. Reaction time fell by 247 ms, 95% CI [206, 287] ms, $t(19) = -11.54$, $p = 5.0 \times 10^{-10}$, $d_z = -2.58$, Wilcoxon $p = 1.9 \times 10^{-6}$, with all 20

faster under C. Key accuracy rose by 0.83 percentage points, 95% CI [0.31, 1.43], $t(19) = 2.82$, $p = 0.0109$, $d_z = 0.63$, remaining near ceiling under both conditions. Participant-level paired differences are shown in Figure 3.4. Condition A agreed with the never-shown fingering on 18.8% of events.

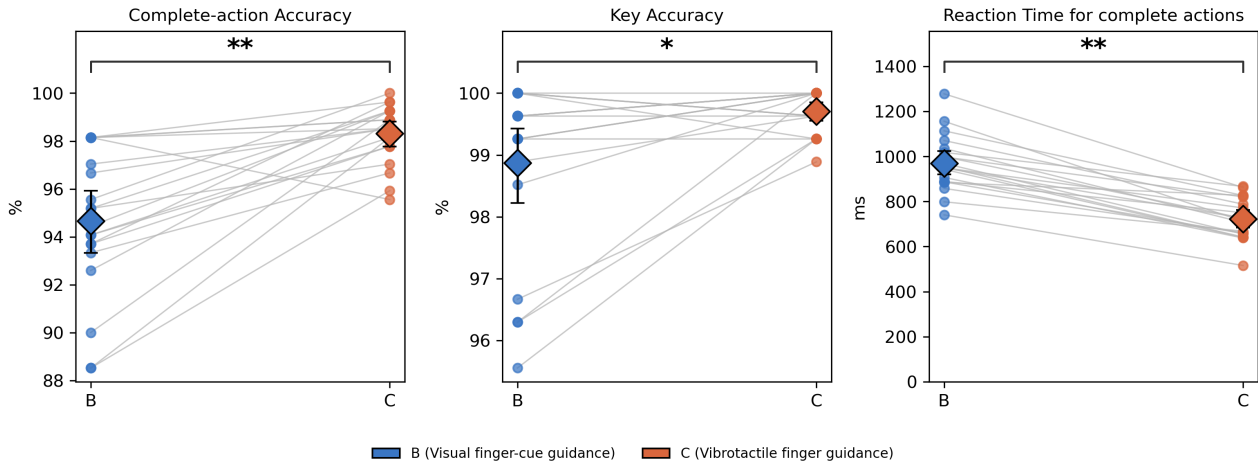


Figure 3.4. Paired visual (B) and vibrotactile (C) finger-guidance outcomes for 20 participants. Grey lines link individual scores, and diamonds show group means with 95% participant-bootstrap CIs. Brackets indicate two-sided paired t -tests, with * $p < 0.05$ and ** $p < 0.01$.

3.2.3 Difficulty

Difficulty increased reaction time, $F(1.57, 29.80) = 31.61$, $p = 2.4 \times 10^{-7}$, $\eta_p^2 = 0.62$, while accuracy stayed within 94.2–95.4% under B and 98.2–98.4% under C. Its interaction with Condition was inconclusive, $F(1.69, 32.17) = 2.76$, $p = 0.086$, $\eta_p^2 = 0.13$. The gap between B and C was 235, 212, and 293 ms across α , β , and γ ; the easy-to-hard change was +58 ms, 95% CI $[-16, +132]$ ms. Thus the data establish the stimulus ordering but neither dependence nor independence of the modality difference from load (Figure 3.5).

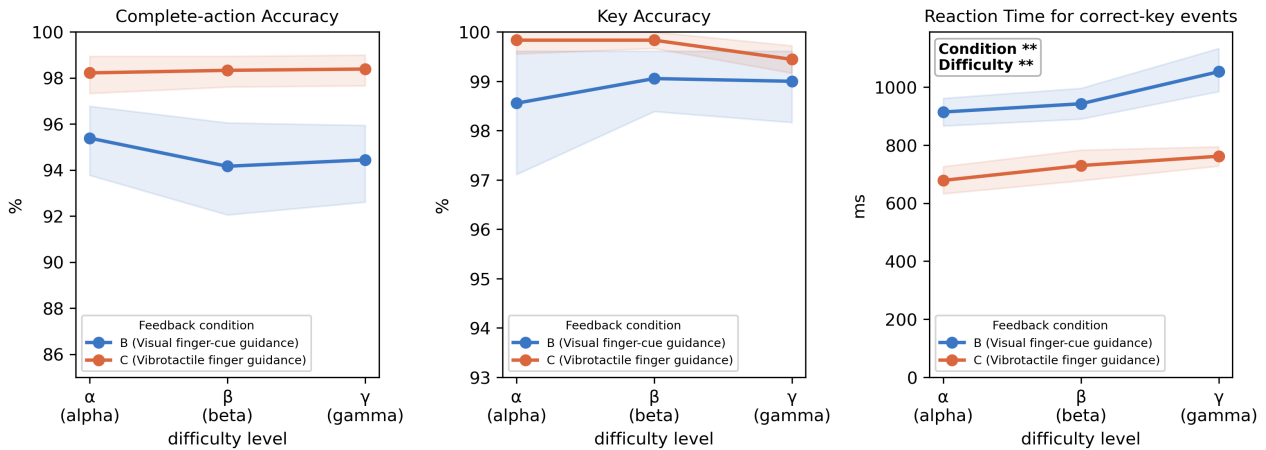


Figure 3.5. Outcomes by difficulty under visual (B) and vibrotactile (C) finger guidance ($n = 20$). Lines show means and 95% participant-bootstrap CIs. The plotted correct-key reaction time is within 1.3 ms of the complete-action time used by ANOVA. The accuracy axes span 85–100% and 93–100%. Stars indicate omnibus reaction-time ANOVA effects, with * $p < 0.05$ and ** $p < 0.01$. Accuracy panels are descriptive.

3.2.4 Finger-Specific Effects

The reaction-time ANOVA returned effects of Condition, $F(1, 19) = 122.22$, $p = 1.0 \times 10^{-9}$, $\eta_p^2 = 0.87$, and Finger, $F(2.52, 47.94) = 26.18$, $p = 2.2 \times 10^{-9}$, $\eta_p^2 = 0.58$, plus a Condition $\times$ Finger interaction, $F(2.43, 46.25) = 7.35$, $p = 0.00087$, $\eta_p^2 = 0.28$. Vibrotactile reductions were largest for the middle and ring fingers (287 and 286 ms), 270 ms for the index, and smallest for the thumb and little finger (186 and 176 ms; Figure 3.6); every plotted 95% participant-level bootstrap CI excluded zero.

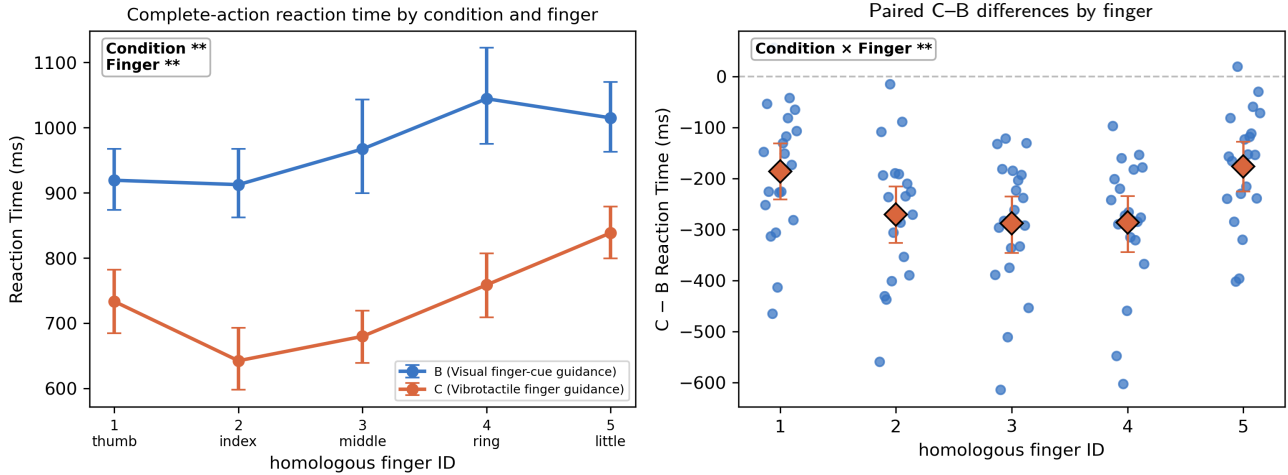


Figure 3.6. Reaction time by finger ($n = 20$). Left shows condition means, right paired $C - B$ differences, with participant points, group diamonds, and 95% participant-bootstrap CIs. Boxed labels name the omnibus ANOVA effects they mark, with $*p < 0.05$ and $**p < 0.01$.

Accuracy became more uniform under C, its per-digit spread falling from 3.9 to 1.7 points (Appendix Figure D.2), whereas reaction-time dispersion increased. The noise-corrected within-participant SD across fingers increased from 71 to 88 ms (Figure 3.7, left). The coefficient of variation rose from 0.085 to 0.130, $t(19) = 4.63$, $p = 0.00018$ (right). The little finger combined a large accuracy gain (4.5 points) with the smallest time gain.

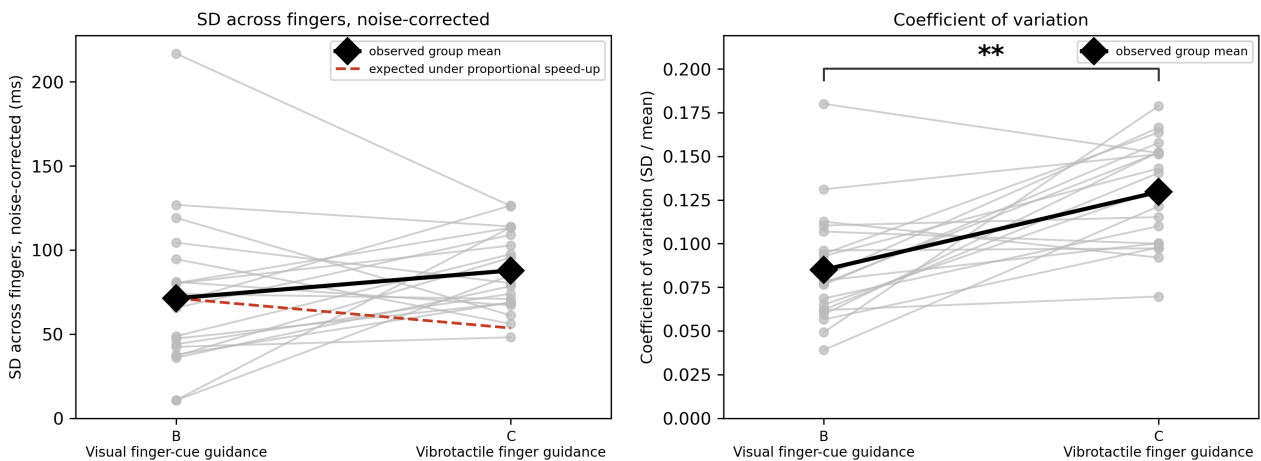


Figure 3.7. Noise-corrected SD (left) and coefficient of variation (right) across fingers ($n = 20$). Grey lines pair participants and diamonds show means. The red dashed line in the left panel shows a proportional-speed-up prediction and is not a confidence interval. The bracket in the right panel indicates the paired t -test, with $*p < 0.05$ and $**p < 0.01$; the left panel is descriptive and unmarked. Definitions are in Appendix C.1.

Complete-action Accuracy reached 1.0 in 66 of 200 participant $\times$ condition $\times$ finger cells. Binomial mixed models accommodated this ceiling (Figure 3.8). The additive model gave an overall C-over-B odds ratio of 3.29, 95% CI [2.19, 4.94], $t(19) = 6.15$, $p = 6.5 \times 10^{-6}$, agreeing with the participant-level paired contrast. The interaction model gave per-digit odds ratios from 2.10 at the thumb to 6.92 at the middle finger. The between-participant SD was 0.46 on the logit scale. The Condition $\times$ Finger interaction was $\chi^2(4) = 10.90$, $p = 0.028$; this exploratory, unadjusted test was interpreted cautiously at $N = 20$. On the automatic pre-screen scores the C over B ordering held at every θ from 0.30 to 0.45 and converged at $\theta = 0.50$.

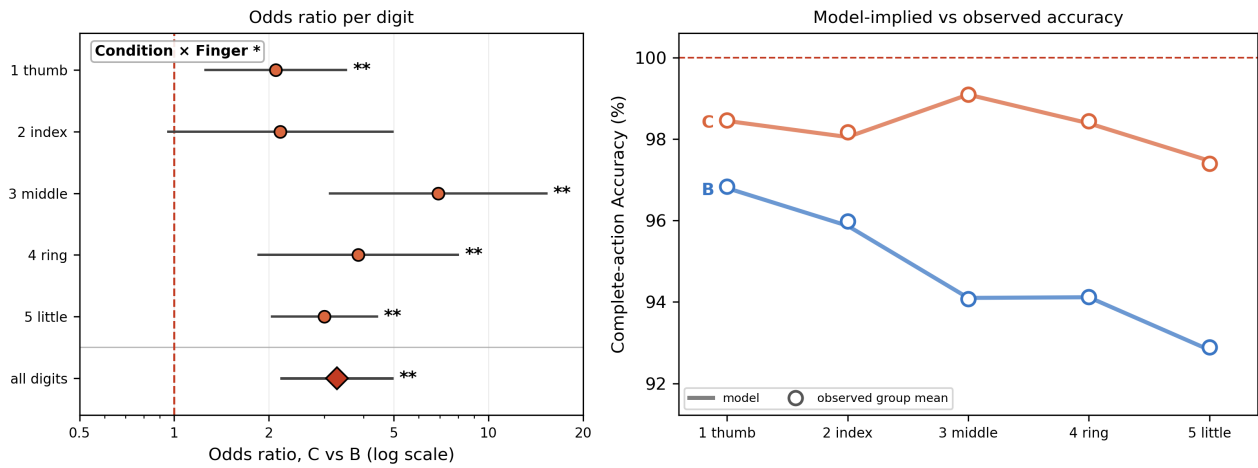


Figure 3.8. Binomial mixed models ($n = 20$). Left shows per-digit odds ratios from the interaction model and the additive overall effect on a logarithmic axis, with 95% participant-clustered intervals and a dashed null line. Stars use clustered t -tests and the interaction likelihood-ratio test, with $*p < 0.05$ and $**p < 0.01$. Digit and interaction tests are exploratory and unadjusted. Right compares predictions from the interaction model (lines) with observed means (circles).

3.2.5 Structure of Finger Errors

Outcome composition separates correct-key events by finger agreement (Figure 3.9). Different-finger events dominated A but fell sharply under B and C; participant rates are in Appendix Figure D.3.

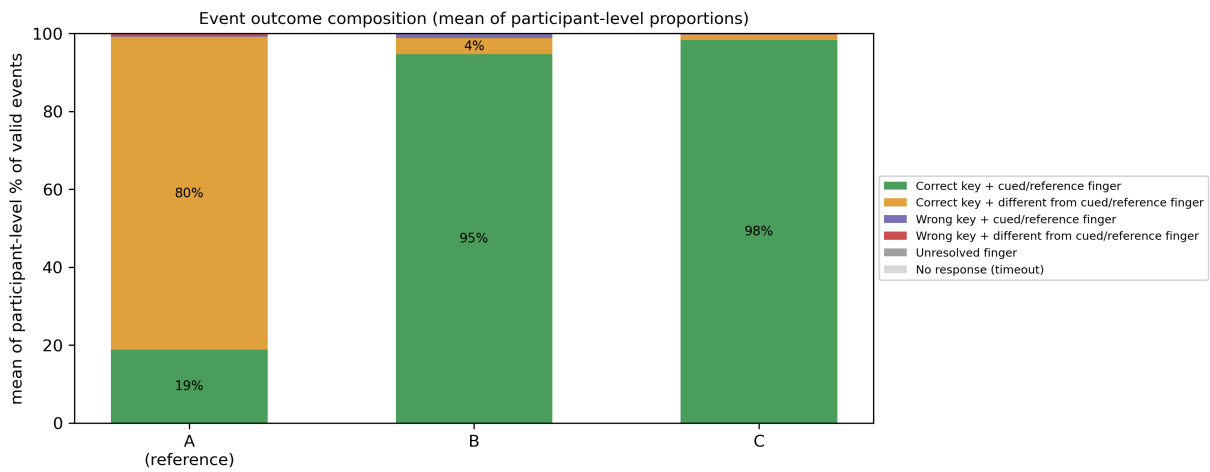


Figure 3.9. Outcome proportions calculated within participant and averaged ($n = 20$), distinguishing key and finger correctness, unresolved events, and timeouts.

Of the 10,799 cued events in Conditions B and C, 10,050 were assigned to the cued finger outright and 443 were off-diagonal events in which it still held at least θ of the mass, adjacent-fingertip ambiguities forgiven by the rule rather than participant errors. The remaining 306 are genuine substitutions, 233 under visual against 73 under vibrotactile guidance (Figure 3.10).

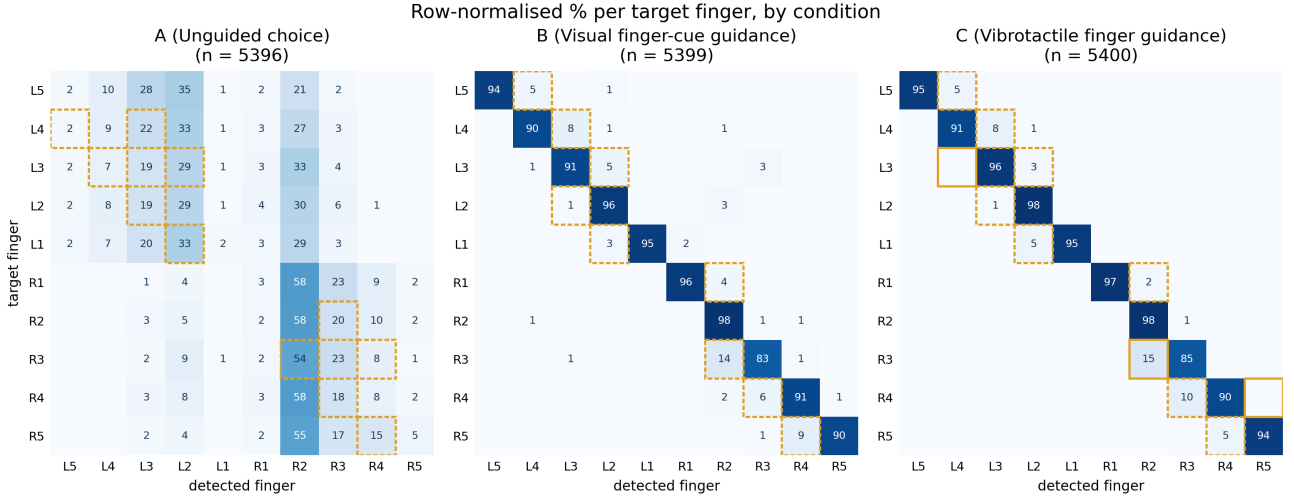


Figure 3.10. Row-normalised target and detected fingers for 16,195 responses from 20 participants, in physical order from L5 to R5. Four timeouts and one carry-over are excluded. Orange dashed and solid outlines mark partly and wholly forgiven near-tie cells. Condition A cues no finger, so its rows describe unguided choice, not accuracy.

Under visual guidance, 67 of 233 substitutions (28.8%) used the wrong hand; under vibrotactile guidance, 2 of 73 (2.7%). Because both conditions contained 5,400 planned events, these counts are directly comparable: same-hand substitutions fell from 166 to 71, cross-hand from 67 to 2.

Within-hand errors were mainly single-position neighbour slips in both conditions (80.1% and 73.2%). Of the visual cross-hand substitutions, 53 (79%) used the homologous digit, against 20% under uniform within-hand choice; the same number were left-hand targets executed with the right hand, most often R2. These pooled counts describe error composition rather than a participant-level test.

The exploratory model estimated stronger hand evidence under C but similar within-hand digit sharpness. Held-out risk enrichment did not exceed the multinomial benchmark; full specification and checks are in Appendix C.2.

3.2.6 Change Within the Session

Reaction time improved with repetition, $F(1.72, 32.65) = 58.22, p = 7.2 \times 10^{-11}, \eta_p^2 = 0.75$, while Completion Accuracy moved little. The Condition $\times$ Repetition interaction was small, $F(1.79, 33.96) = 0.13, p = 0.853, \eta_p^2 = 0.01$. The gap between B and C changed from 253 ms at the first repetition to 239 ms at the third, a change of -14 ms, 95% CI $[-70, +43]$ ms (Figure 3.11; difficulty-aligned all-condition trajectories are in Appendix Figure D.4).

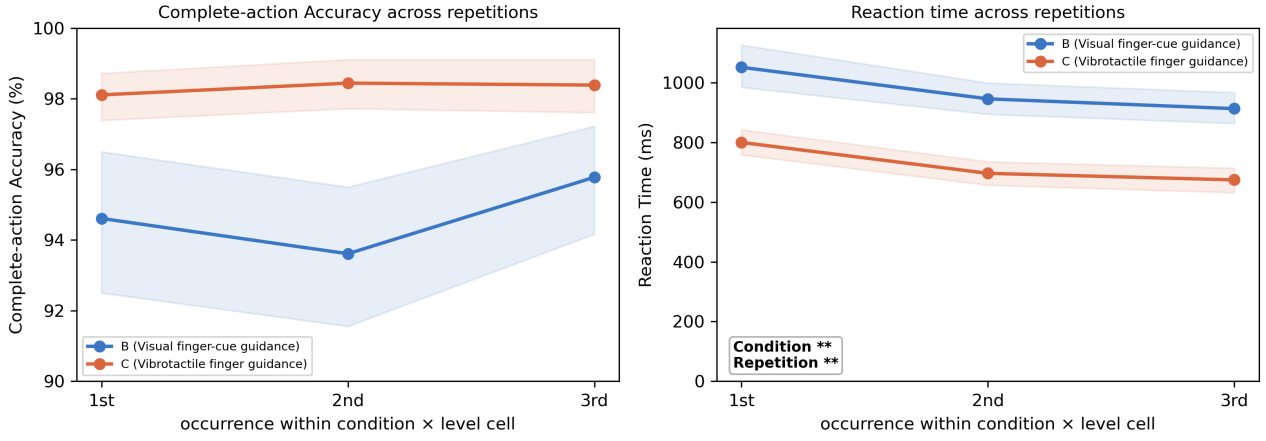


Figure 3.11. Complete-action Accuracy and reaction time over three occurrences per condition and difficulty ($n = 20$). Bands show 95% participant-bootstrap CIs; the accuracy axis spans 90–100% and the reaction-time axis starts at zero. Stars indicate omnibus reaction-time ANOVA effects, with $*p < 0.05$ and $**p < 0.01$. Accuracy is descriptive.

3.2.7 Unguided Finger Choice in Condition A

In A, four digits absorbed 85.2% of presses, R2 alone 42.5% (Figure 3.12, left). Pooling resolved presses across each participant’s nine A trials gave 4.81 ± 1.30 effective fingers (mean $\pm$ SD, not plotted).

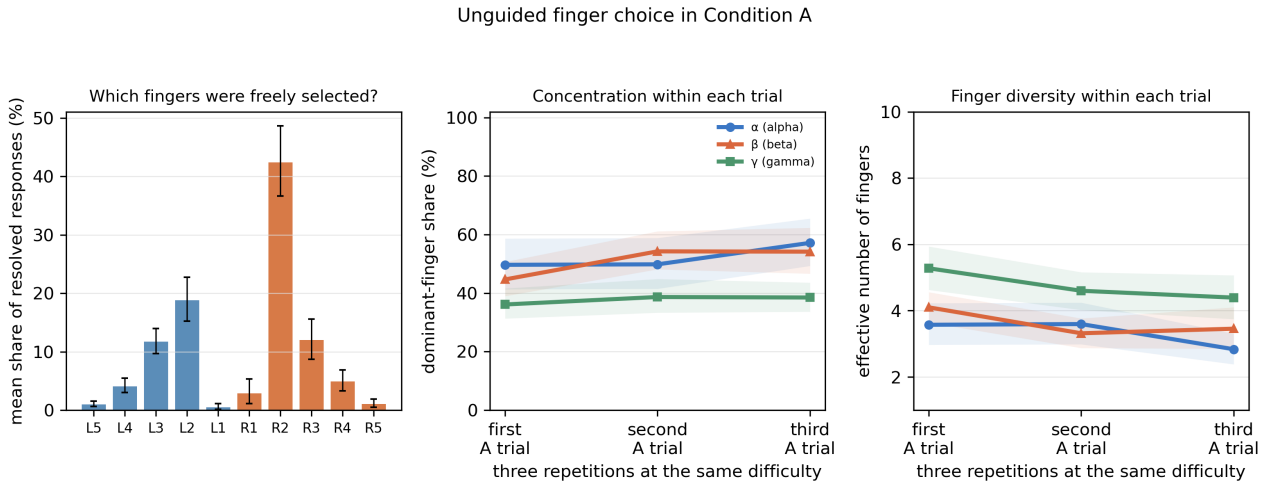


Figure 3.12. Condition-A finger use pooled within participant (left); trial-wise dominant-digit share (middle) and effective finger count, $\exp(-\sum_f p_f \log p_f)$ for finger-use proportions p_f (right), by difficulty and repetition ($n = 20$). Means and 95% participant-bootstrap CIs; left-panel colours distinguish hands.

Right-hand use exceeded the near-balanced generated fingering by 13.8 percentage points, $t(19) = 6.55$, $p = 2.8 \times 10^{-6}$; the excess was concentrated in right-handed participants (Appendix Figure D.5).

Same-hand finger-transition distance was roughly half the generated value at every level, while key travel did not change consistently (Figure 3.13).

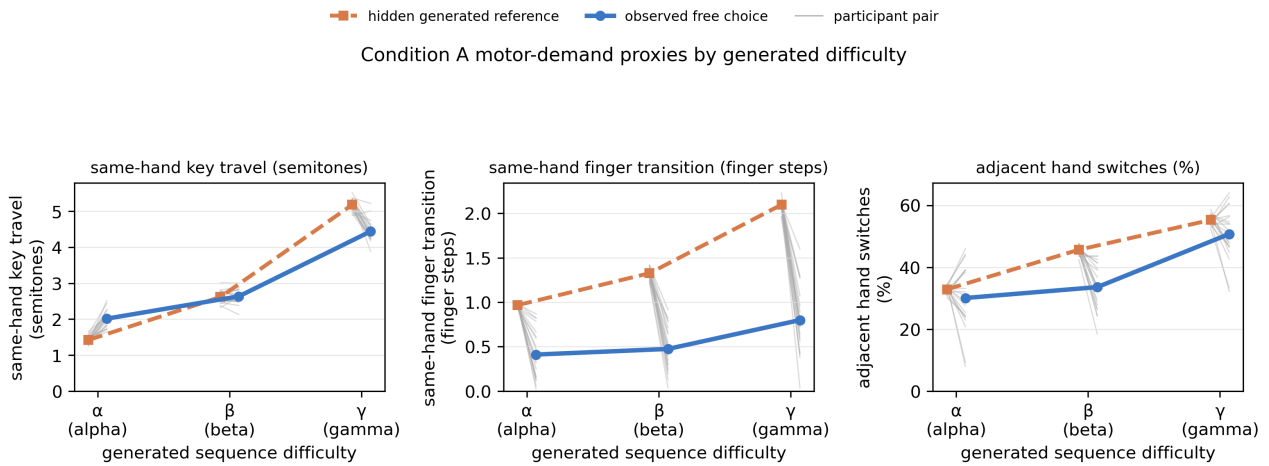


Figure 3.13. Motor-demand proxies in A by generated difficulty ($n = 20$). Dashed lines show generated references, solid lines observed means, and thin lines participant pairs.

3.3 Tele-training Deployment

All 1,000 transport probes arrived, with median round-trip time 36.6 ms; 35.5% exceeded 100 ms. Across 244 student-side cues, dispatch took a median 1.1 ms; among 160 dual-modality cues, software skew between visual and haptic output stayed within 0.47 ms. Component-based note-on-to-cue budgets were approximately 58 ms using typical values and 170 ms using component p95 estimates; end-to-end percentiles were not measured. Local alignment was tight, but network tail latency remains relevant to synchronous performance. Single-operator demonstrations verified delivery without estimating a remote behavioural effect (Appendices B and D).

Chapter 4

Discussion

4.1 What the Experiment Establishes

With finger information and difficulty matched, vibrotactile cues reduced reaction time by 247 ms and increased Complete-action Accuracy by 3.65 percentage points relative to visual cues, answering RQ1. Digit-dependent effects and fewer wrong-hand substitutions answer RQ2. Ten nail-mounted actuators, a reviewed camera and MIDI chain, and validated difficulty delivered and scored finger-specific guidance with the fingerpad free. Changing cue modality while preserving finger information changed both action time and the selected effector.

4.2 Relation to Vibrotactile Guidance of the Hand

Coscia and Al Borno found lower timing error with vibrotactile than visual stimulation in novice piano learning [10]. Their vibration specified the finger, but vision only the key. Here both specified the digit and shared key illumination. Their non-significant note-accuracy estimate favoured vision and was measured after withdrawal rather than during guidance.

Comparing vibration, electrical muscle stimulation, and a linkage exoskeleton, *Hapticus* found no modality dominant across outcomes [11]. These devices differ in physical assistance and sensory channel. The present cues specified actions without imposing movement, supporting stimulation-site encoding of discrete finger identity rather than movement magnitude. LRA selection before recruitment limited hardware delay relative to behavioural effects; bench tests did not calibrate visual–tactile perception.

4.3 Finger Selection and Alternative Explanations

The digit and error patterns are consistent with reduced finger-selection demands. The little finger’s large accuracy but smallest time gain is informative because it is intermediately coupled yet actively constrained: a cue can resolve its identity without removing movement cost. Index accuracy was already near ceiling, leaving its benefit mainly in time. Complete-action Accuracy became more uniform while reaction-time dispersion increased, consistent with removal of a cue-dependent selection cost while digit-specific movement costs remain. Cue-to-keypress time records both.

Reaction-time gains were largest for the index, middle, and ring fingers, which Lang and Schieber found least mechanically independent [1]; their active-control measure constrained the ring and little fingers most. This post hoc rank correspondence was untested, but the literature-derived ordering was independent of the analysis. Furuya and Altenmüller group the index differently [2], further limiting the account. Smaller modulation in trained pianists with more equal individuation remains a prediction.

Wrong-hand substitutions fell from 67 to 2 under vibrotactile guidance, versus 166 to 71 within-hand. The selective 34-fold versus 2.3-fold reduction and 1.6-fold variation in time benefit across digits favour a contribution from spatially specific cueing, without excluding general salience. Most visual wrong-hand errors preserved the homologous digit, 79% against 20% under uniform within-hand choice, consistent with difficulty assigning a finger number to the correct hand. They often recruited R2, also dominant in A, suggesting habitual choice. Persistent within-hand neighbour slips fit imprecise execution. Attention and action planning could also contribute; pooled C errors are too sparse for participant-level inference. The exploratory model did not outperform the multinomial benchmark and remains descriptive (Appendix C.2).

Key accuracy also favoured C by 0.83 percentage points ($p = 0.0109$) despite identical key illumination, so the benefit extended beyond finger correctness. This permits contributions from general salience, engagement, or reduced competition for visual attention, without distinguishing them. A's 99.1% versus B's 98.9% is directionally consistent with an attentional cost, but this descriptive comparison also changes the finger-selection requirement (Table 3.1). The small display kept cues, keys, and hands in one field of view (Section 2.3.1), but eye movements were unrecorded. Gaze or attentional demands could explain part of the time gap; a fixed gaze penalty alone would not predict the finger interaction or homologous wrong-hand errors. The design cannot separate these costs from symbolic-to-body translation.

Physical onset was unmatched. The LRA began moving about 6.1 ms after command, but display latency and perceptual detection were unmeasured. Detection may occur during actuator rise, so settling time is not a known tactile disadvantage. This uncertainty affects interpretation of the reaction-time contrast's magnitude; the observed accuracy and error composition do not depend on where timing begins. Detection-latency measurements are needed to determine the correction's sign and size.

4.4 Difficulty, Practice, and Unguided Choice

Difficulty increased reaction time, but its interaction with Condition was inconclusive ($p = 0.086$, $\eta_p^2 = 0.13$). The easy-to-hard change in the modality gap was +58 ms, 95% CI $[-16, +132]$ ms. This spans little change and substantial widening; a moderate interaction remains unresolved at $n = 20$, establishing neither dependence on nor independence from load.

Practice produced a large effect ($\eta_p^2 = 0.75$), while the gap changed by -14 ms, 95% CI $[-70, +43]$ ms, across three repetitions. This excludes rapid convergence on the scale of the 247 ms mean gap without establishing equivalence, providing bounded evidence against initial display unfamiliarity as the sole explanation. Automatisations over days or weeks remains untested (Figure 3.11).

Condition A constrained only the key and remains descriptive. Its 99.1% key accuracy coexisted with 18.8% hidden-fingering agreement, showing why MIDI-only evaluation misses finger selection. It reveals a four-digit strategy rather than expert fingering, with shorter finger transitions but no consistent reduction in key travel (Figures 3.12 and 3.13).

4.5 Guided Performance and Learning

All outcomes were measured with cues present. The guidance hypothesis distinguishes improved acquisition performance from lasting learning because reliance on feedback may impair performance after withdrawal [6, 7]. Specificity of learning likewise describes incorporation of useful afferent information into the task, explaining dependence on it without predicting superior retention after tactile guidance [7]. The present 247 ms advantage is consequently not evidence of better learning. Whether a body-referenced movement representation persists after cue removal is a testable hypothesis for withdrawal and delayed-retention tests.

4.6 Limitations

Sample and task. Twenty participants from one institution, mostly inexperienced and including only one woman, limit generalisation across expertise and sex. Repeated events improve individual measurement without increasing independent sample size. Single-key, self-paced responses on white keys with muted audio exclude polyphony, expressive timing, and external tempo [26].

Finger labels. Every scored event was reviewed, but a single reviewer provides no inter-rater reliability estimate. Rater bias could affect accuracy, error composition, and inclusion in complete-action reaction time, although MIDI timestamps themselves are unaffected. Correct-key reaction time produced an almost identical condition difference, supporting robustness to that inclusion rule. The threshold forgives near-ties. Automatic scores retained the C-over-B ordering for $\theta = 0.30$ to 0.45 and converged at 0.50 ; absolute accuracy remains conditional on the scoring criterion.

Ceiling and multiplicity. Sixty-six of 200 accuracy cells were perfect. The binomial sensitivity analysis accommodated this ceiling and agreed with the paired contrast, but per-digit rankings changed between percentage-point and odds-ratio scales, because at these high baselines the same odds ratio yields fewer percentage points; neither scale uniquely ranks digit benefit. Ceiling compression leaves the size and direction of any hidden differences unresolved. Difficulty, repetition, and per-digit accuracy analyses remain secondary or exploratory with unadjusted tests, and neither secondary interaction supports a confirmed null.

Tele-training. Local software alignment was tight but transport was variable, and the single-operator demonstrations lacked counterbalancing and manual review. They establish delivery, not a remote behavioural effect; more than a third of probes exceeded 100 ms, which would matter whenever teacher and student share a beat. Frame-by-frame review is unavailable in a live session, so the student receives the automatic pre-screen label. One reviewer, with no second rater, changed 4.50% of those labels in the main study (Section 3.2.1), so a similar share of remote cues would name a finger that review would have altered. Most such events were near-ties between adjacent fingertips, so the likely miscue is a neighbouring digit rather than an arbitrary one. Unattended remote use needs teacher-side confirmation or a detector validated against reviewed labels.

4.7 Future Work

First, test withdrawal and delayed retention across sessions, powering the study on participants. Second, combine eye tracking, perceptual-onset matching, a visual cue spatially co-located with the hand, expert participants, and an independent label rater to separate gaze, mapping, individuation, and measurement effects. Third, conduct a counterbalanced, manually reviewed remote comparison while controlling network tail latency. External tempo and other finger-assignment tasks would then test transfer beyond this paradigm.

For biomedical engineering, separating key success from finger choice offers a test bed for fine-motor guidance while leaving the fingerpad free. Training and rehabilitation, including remote delivery, are potential applications. Translation requires testing perceptual accessibility and functional transfer in people with motor or sensory impairments; this single-session study establishes neither rehabilitation efficacy nor equitable access.

Chapter 5

Conclusion

The platform met the aim of delivering and measuring nail-mounted finger-specific guidance with the fingerpad free and supported tele-training. Within one session, vibrotactile guidance reduced reaction time by 247 ms and improved Complete-action Accuracy by 3.65 percentage points compared with matched visual cues. Digit-dependent benefits and fewer cross-hand substitutions are consistent with reduced finger-selection demands, although selection and execution were not measured separately. For biomedical engineering, this offers a test bed for immediate fine-motor guidance; retention after cue withdrawal, remote behavioural benefit, and rehabilitation efficacy remain untested.

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Appendix A

Stimulus Pool Provenance and Validation

The formal stimulus pool was generated and locked with

```
generation_seed = 1799551287,
```

nine sequences per difficulty bin, on the calibrated profile spanning MIDI notes 48–72. Saved sequence names carry the seed as their batch prefix (e.g. 1799551287- α -1). Rerunning the generator with the same profile, note range, family size, and seed at the code state recorded in Table A.1 reproduces the pool exactly, so the 27 complete 30-event action lists are not reprinted here: the stored pool itself, not a printed transcript, is what a replication should be checked against. Table A.1 gives that provenance, and Table A.2 summarises the per-level validation statistics behind the **VALID** conclusion visualised in Figure A.1.

Table A.1. Provenance of the locked stimulus pool: the seed, code state, and stored locations from which the 27 sequences can be regenerated and checked.

Item	Value
Generation seed	1799551287
Pool	3 difficulty levels $\times$ 9 sequences = 27
Sequence length	30 cue events, one complete action each
Note range	MIDI 48–72, white keys only (15-note pool)
Generator module	main/app/sequence_generator.py
Validation module	main/app/stimulus_validation.py
Generator last changed	commit 6d4d2c4 (2026-08-03)
Locked stimuli	main/data/sequence/1799551287- $\{\alpha, \beta, \gamma\}$ - $\{1..9\}$ /
Validation record	main/data/sequence_validation/1799551287/

Table A.2. Per-level validation summary of the locked pools: median [Q1, Q3] per component ($n = 9$ per level), adjacent-pair violation rate (limit 10%), and verdict. Distances are in semitones on the calibrated profile.

Component	Expected	α	β	γ	Viol.	Verdict
d_{seq}^m	rise	2.62 [2.48, 2.69]	3.38 [3.00, 3.55]	6.97 [6.59, 7.59]	1.2%	monotonic
d_m^m	rise	1.46 [1.39, 1.54]	2.89 [2.36, 2.96]	5.11 [4.89, 5.46]	0.0%	monotonic
d_{95}^m	rise	2.00 [2.00, 2.00]	5.00 [4.00, 5.00]	9.65 [9.00, 10.0]	0.0%	monotonic
d_f^m	rise	1.00 [0.96, 1.00]	1.29 [1.25, 1.43]	2.14 [2.00, 2.18]	0.0%	monotonic
R_L	rise	3.00 [3.00, 6.00]	8.00 [7.00, 10.0]	14.0 [12.0, 14.0]	3.7%	monotonic
R_R	rise	5.00 [3.00, 5.00]	7.00 [5.00, 7.00]	15.0 [14.0, 15.0]	9.9%	monotonic
H_{norm}	rise	0.27 [0.27, 0.28]	0.51 [0.50, 0.51]	0.69 [0.68, 0.71]	0.0%	monotonic
V_{trans}	rise	0.10 [0.10, 0.10]	0.24 [0.21, 0.24]	0.31 [0.31, 0.38]	0.0%	monotonic
P_{pred}	fall	0.17 [0.17, 0.21]	0.10 [0.07, 0.10]	0.07 [0.07, 0.07]	0.0%	monotonic
A_h	rise	0.34 [0.31, 0.34]	0.45 [0.45, 0.48]	0.55 [0.55, 0.59]	0.0%	monotonic
H_{hand}	diagnostic	0.95 [0.94, 0.95]	0.99 [0.99, 0.99]	0.99 [0.98, 1.00]	—	reported only
B_h	matching	0.87 [0.87, 0.87]	0.93 [0.93, 0.93]	0.93 [0.93, 0.93]	—	above floors
O_{LR}	rise	0.00 [0.00, 0.00]	0.07 [0.06, 0.08]	0.26 [0.25, 0.27]	0.0%	monotonic
X_f	cross limits	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.13 [0.10, 0.20]	—	within limits
X_e	cross limits	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	3.00 [3.00, 3.00]	—	within limits

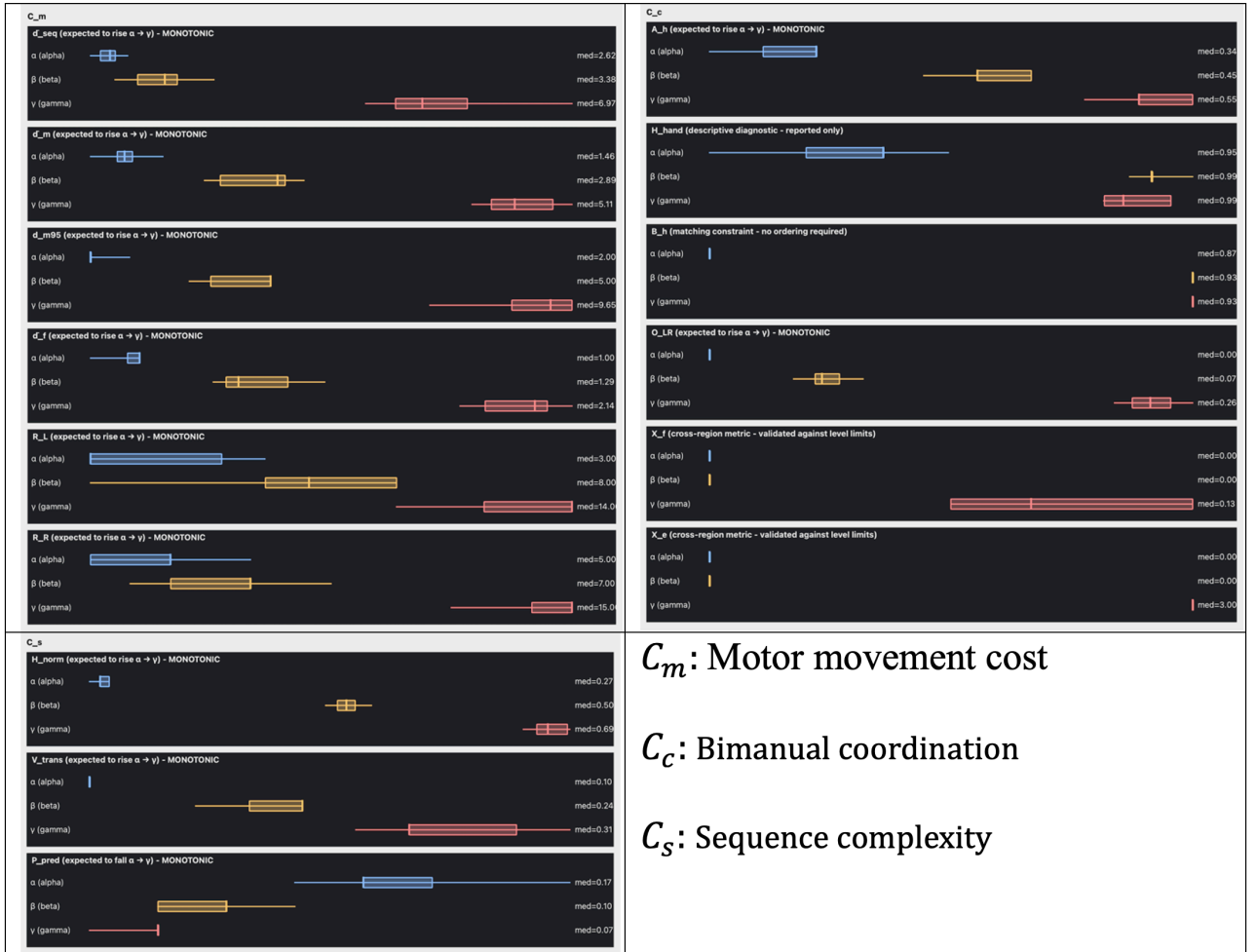


Figure A.1. Locked-pool component distributions by level ($n = 9$): motor movement cost C_m , bimanual coordination C_c , and sequence complexity C_s . See Table A.2 for validation.

All sequences use both hands and pass all checks: no repeated three-event chunk, no hand above 60% share, no finger run longer than three, and consistent fingering on repeated keys.

Appendix B

Generator, Synchronisation, and Matching Detail

This appendix records the implementation detail omitted from Chapter 2: the bench-validation rig and drive sweeps, the stimulus generator’s internals and difficulty components, the video–MIDI synchronisation self-checks, the calibration of the finger-matching decision rule, and the tele-training measurement protocol. The Python entry points are:

- generation: `main/app/sequence_generator.py`;
- validation: `main/app/stimulus_validation.py`;
- anchoring: `main/app/sync_led.py`; and
- matching: `main/app/finger_matching.py`.

This appendix and that code are the reproduction reference for the locked pools of [Appendix A](#).

B.1 Bench Validation Rig and Drive Sweeps

All three bench experiments of Section 2.2 drove the actuator through the experimental firmware and scored vibration from a coupled LIS3DH streaming three axes at 1.344 kHz over a shared SPI bus, intensity being the demeaned three-axis vector RMS over N samples,

$$a_{\text{rms}} = \sqrt{\frac{1}{N} \sum_{i=1}^N [(x_i - \bar{x})^2 + (y_i - \bar{y})^2 + (z_i - \bar{z})^2]}.$$

Per-axis demeaning removes gravity, static offset, and mounting orientation together; a $\sqrt{x^2 + y^2 + z^2}$ magnitude is only second-order sensitive to vibration perpendicular to gravity and so mixes rig pose into every reported intensity.

Drive parameters. Frequency was swept from 100–350 Hz and re-swept at 1 Hz resolution around the peak, locating the mounted resonance at 224 Hz. The firmware’s PWM amplitude parameter, amp_{PWM} , was then swept from 4 to 128 at that resonance. It is an 8-bit setting mapped to duty cycle as $D = \text{amp}_{\text{PWM}}/255$; the driver therefore applies a rectangular 0–9 V pulse train, with high time D/f , rather than a sinusoidal voltage. The dashed $\sin(\pi D)$ curve in Figure B.1d is only a scaled model of the PWM fundamental at the drive frequency, not the applied waveform. The selected 25.1%-duty point produced 2.68 m/s^2 , the measured value closest to the provisional target of 2.7 m/s^2 within the 2.4–3.0 m/s^2 band; it was not the strongest setting. The band rescales the drive range this project had previously used onto the demeaned vector-RMS metric. It is not an independent psy-

chophysical calibration, so the selection fixes a reproducible operating point rather than a perceptually justified intensity.

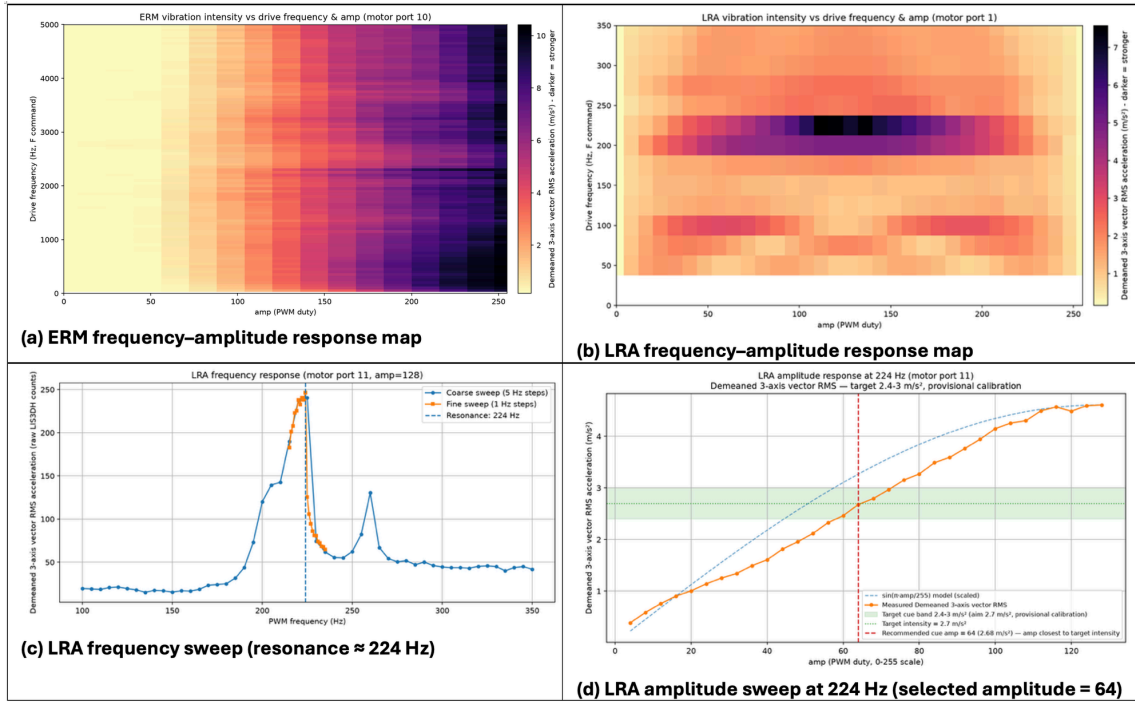


Figure B.1. Actuator characterisation and LRA parameter selection, one mounted assembly per actuator. (a, b) Vibration intensity over the frequency–amplitude plane for the ERM and LRA. (c) LRA frequency sweep locating the mounted resonance. (d) PWM–amplitude sweep at 224 Hz with the target cue band and selected PWM amplitude parameter.

Actuator comparison. Actuator and accelerometer were rigidly coupled and suspended freely, since rigid bench mounting damps the calibrated cue below reliable onset detection. Each actuator ran under its own study configuration — the LRA at its calibrated 224 Hz resonance, and the ERM on a 1 kHz PWM carrier at 19.6% duty because a DC motor cannot start under low-frequency chopping. Onset was estimated by CUSUM change-point detection [29], which does not require a fixed amplitude criterion, and settling time was scored separately as the interval from detected onset to entry into the steady band, so that when the actuator began to move and when it reached working amplitude are reported as distinct quantities.

Attachment. Each adhesive was characterised over three independent mounts of five trials at the fixed drive setting, with the mount as the averaging unit, since re-applying an adhesive is itself a source of variation. Reported quantities are the mean of per-mount means, the between-mount standard deviation, a robust peak, and total harmonic distortion (Table B.1). Here the actuator reaches the accelerometer through the adhesive under test, so these levels index transmission through each bond and run several times higher than the free-suspended actuator output of Figure 3.2; they rank the three adhesives against one another and are not on the same absolute scale as the 2.68 m/s² drive-calibration point.

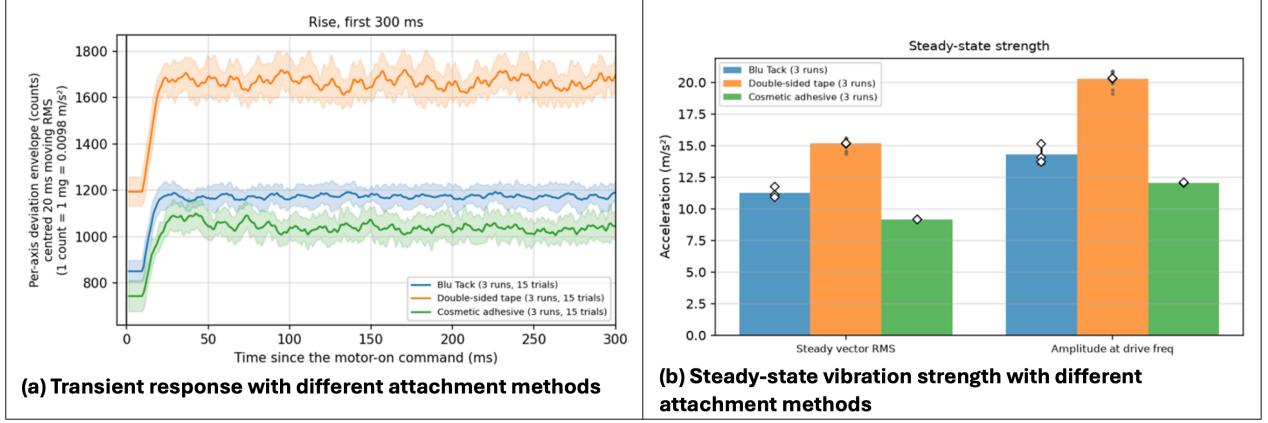


Figure B.2. Vibration transfer by attachment, three independent mounts of five trials per adhesive: (a) rise over the first 300 ms; (b) steady-state vector RMS and amplitude at the drive frequency. Amplitudes are transmission through the adhesive bond, not the free-actuator output of Figure 3.2.

Table B.1. Steady vibration transfer from three mounts of five trials per adhesive. Values are means of mount means, with between-mount SD in parentheses.

Method	Mounts	Vector RMS (m/s ²)	Robust peak (m/s ²)	THD (%)	Relative (dB)
Blu Tack	3	11.27 (0.45)	17.24	23.9	0.00
Double-sided tape	3	15.18 (0.04)	23.99	11.6	+2.58
Cosmetic adhesive	3	9.14 (0.01)	15.81	17.2	-1.82

B.2 Difficulty Components

Table B.2. Level-defining constraints; ranges using the note span S are normalised by it. All levels use both hands. Overlap ratio r is 0, 0.15, 0.30 for α , β , γ ; only γ permits crossing.

Level	Motor and sequence constraints	Coordination constraints
α	$\bar{d}^{\text{seq}}/S \leq 0.12$, $\bar{d}^m/S \leq 0.10$, $d_{95}^m/S \leq 0.16$, $\bar{d}^f \leq 1.0$, $R_h/S \leq 0.30$, $0.15 \leq H_{\text{norm}} \leq 0.35$.	$0.15 \leq A_h \leq 0.35$, $B_h \geq 0.65$, $O_{LR} = 0$, $X_f = 0$, $X_e = 0$.
β	$0.08 < \bar{d}^{\text{seq}}/S \leq 0.22$, $0.08 < \bar{d}^m/S \leq 0.20$, $d_{95}^m/S \leq 0.30$, $\bar{d}^f \leq 2.0$, $R_h/S \leq 0.45$, $0.40 \leq H_{\text{norm}} \leq 0.65$.	$0.30 \leq A_h \leq 0.55$, $B_h \geq 0.70$, $0 < O_{LR} \leq 0.15$, $X_f = 0$.
γ	$0.15 < \bar{d}^{\text{seq}}/S \leq 0.38$, $0.15 < \bar{d}^m/S \leq 0.35$, $d_{95}^m/S \leq 0.45$, $\bar{d}^f \leq 3.0$, $R_h/S \leq 0.65$, $0.65 \leq H_{\text{norm}} \leq 0.90$.	$0.50 \leq A_h \leq 0.80$, $B_h \geq 0.75$, $0.10 \leq O_{LR} \leq 0.30$, $0 < X_f \leq 0.20$, $X_e/S \leq 0.20$.

Motor cost C_m collects, in semitones, the mean displacement between consecutive events $\bar{d}^{\text{seq}}$, the mean and 95th-percentile same-hand key displacement $\bar{d}^m$ and d_{95}^m , the mean finger-transition distance $\bar{d}^f$, and the per-hand range R_h . Sequence complexity C_s classifies each transition by hand switch, binned displacement, and binned finger jump, and summarises the resulting distribution by normalised class entropy H_{norm} over $K = 16$ classes, class variety V_{trans} , and predictability $P_{\text{pred}} = \max_{\delta} \hat{p}(\delta)$. The last is estimated over *relative* moves $\delta_t = (I(h_{t+1} \neq h_t), f_{t+1} - f_t, m_{t+1} - m_t)$ rather than over absolute actions, since a count-based estimate on 30 events would give almost every once-observed action probability 1 and so score the most varied sequences as

the most predictable. Coordination complexity C_c comprises hand alternation A_h , hand balance B_h , the overlap O_{LR} of the hands' used note intervals, and the frequency X_f and extent X_e of cross-region events.

B.3 Hand Regions and Cross-Region Rules

Given the effective note bounds $k_{\min}$, $k_{\max}$, span $S = k_{\max} - k_{\min}$, and midpoint $k_{\text{mid}} = k_{\min} + S/2$, each level defines hand operating regions from an overlap ratio r :

$$R_L(r) = \left[k_{\min}, k_{\text{mid}} + \frac{rS}{2} \right], \quad R_R(r) = \left[k_{\text{mid}} - \frac{rS}{2}, k_{\max} \right],$$

with $r = 0$ for α (strict split, no crossing), $r = 0.15$ for β (small shared middle band), and $r = 0.30$ for γ . When the experimenter narrows the note range, $k_{\min}$, $k_{\max}$, and S follow the selected range, so hand regions, midpoint, and every span-normalised constraint describe the keyboard area actually in use; the effective bounds are stored in each saved sequence's metadata so that later recomputation uses the same span as generation.

A cross-region event occurs when the left hand is assigned above k_{mid} or the right hand below it:

$$X_t = I((h_t = L \wedge m_t > k_{\text{mid}}) \vee (h_t = R \wedge m_t < k_{\text{mid}})),$$

with frequency $X_f = \frac{1}{T} \sum_t X_t$ and extent $X_e = \max_t X_t |m_t - k_{\text{mid}}|$. Only γ permits $X_t = 1$, and then only when all ergonomic checks pass: $X_e/S \leq 0.20$; no more than two consecutive cross-region events; no jump into or out of a crossing greater than $0.45S$; and, in a crossed adjacent opposite-hand pair (the left-hand note lying above the right-hand note), the hand on the wrong side of the midpoint must use a thumb-side finger (1–3), since hand-over crossing with the ring or little finger is not a plausible movement.

Two further coordination components are defined as follows. Hand alternation is $A_h = \frac{1}{T-1} \sum_{t=2}^T I(h_t \neq h_{t-1})$. The hands' used note intervals are $U_h = [\min_{t \in Q_h} m_t, \max_{t \in Q_h} m_t]$ over each hand's active indices Q_h , and their overlap is $O_{LR} = |U_L \cap U_R| / |U_L \cup U_R|$, with interval sizes counted as the number of semitone positions covered inclusive of both endpoints. That convention is what makes the β -level combination of $X_f = 0$ with $O_{LR} > 0$ satisfiable: both hands must reach the midpoint note, but neither may enter the opposite register.

B.4 Candidate Construction and Rejection Rules

The generator samples a target level, then a left/right hand-label sequence satisfying that level's A_h and B_h constraints, then enumerates legal continuations satisfying the same-hand movement constraints, the hand-region rule, the overlap and cross-region constraints, the requirement that both hands appear, and the rule that each event contains exactly one action. If no legal continuation exists the candidate is discarded and generation restarts.

Legal continuations are sampled with level-specific proposal heuristics — persistence toward the dominant transition class and the dominant complete relative move, note locality around hand switches, and level-typical finger-jump windows — because several constrained bands, notably the α -level H_{norm} range, are reached with negligible probability by uniform sampling over 29 transitions. These heuristics shape only the proposal distribution: acceptance of a finished candidate is decided solely by the constraint table and the rejection rules, so no sequence outside the intended level ranges can be admitted.

A completed candidate is rejected if any level-defining component of C_m , C_s , or C_c lies outside the target level's range, if any event contains more than one action, if it contains repeated identical three-event chunks, if one hand accounts for more than 60% of events, or if the same hand uses one finger for more than three

consecutive occurrences. Consecutive repetitions of the same key by the same hand preserve the same finger assignment, avoiding unrealistic repeated-key finger substitutions. Hand-transition entropy is logged but never used for rejection.

B.5 Family Matching Tolerances

Accepted candidates are deduplicated on their exact event lists, then connected in a compatibility graph in which an edge joins two candidates satisfying every pairwise tolerance in Table B.3; a greedy clique heuristic extracts a mutually compatible family of the requested size. If no family can be matched within tolerance, the generator reports an error for that difficulty bin rather than relaxing the constraint silently, and the experimenter may widen the note range, relax the tolerances, or request a smaller pool.

Table B.3. Pairwise matching tolerances for sequences within the same level pool.

Statistic	Tolerance
H_{norm}	± 0.05
$\bar{d}^m / S$	± 0.04
A_h	± 0.08
B_h	± 0.10
O_{LR}	± 0.05

Saved stimuli are written as a metadata file and a fingering file per sequence, in the same layout used for recorded material, so the playback and cue-delivery tools load generated sequences without special-case handling. The stored target note is accompanied by the resolved physical key identifier from the active profile’s MIDI mapping, so cue delivery addresses the correct calibrated key without relying on a hard-coded key index. A read-only metrics viewer recomputes every component of C_m , C_s , and C_c for saved sequences using the same functions as the generator, so metrics shown during generation and metrics reported later are directly comparable.

B.6 Synchronisation Anchor Self-Checks

The per-trial video–MIDI anchor of Section 2.3.1 is accepted only when two independent self-checks pass. First, the pulse length recovered from the video must match the commanded flash duration, computed from the saved LED-on and LED-off timestamps, within a fixed tolerance; a bright region of the wrong duration is not the mark. Second, the trial offset implied by the rising edge and the offset implied by the falling edge must agree, which fails if either edge was mis-localised by motion or exposure change.

Detection proceeds on the mean brightness of one calibrated key’s pixel region over the opening seconds of the video. A sliding box matched filter of exactly the commanded width selects the window that is bright inside and dark immediately before and after; the search is restricted to ± 0.5 s around the software-expected flash time so that later cue LEDs on the same key cannot be mistaken for the mark, and both edges are then refined on the frame-to-frame brightness derivative. Trials failing either check fall back to the recording start-timestamp estimate and are flagged for frame-level inspection in the review tool, which also serves as a spot audit of the automatic anchors. Each trial stores the anchor it actually used, so any analysis can be traced to its alignment evidence.

B.7 Finger-Matching Decision Rule

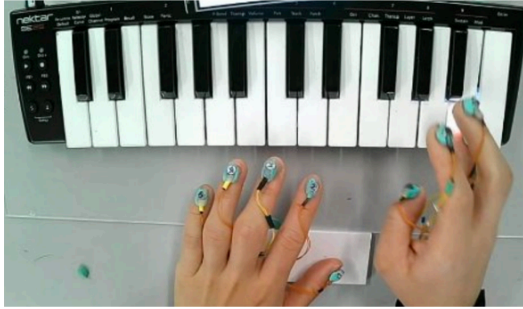
Section 2.3.3 gives the reason for a graded rather than a nearest-fingertip decision; this section records the parameter values it produces. The temperature $\tau = 25$ px is approximately half a white-key width at the working resolution of 1280×720 , so ambiguity is calibrated in units of the response surface: two fingertips within half a key of each other share the mass almost evenly, while a fingertip a full key away is outweighed by a factor of about $e^2 \approx 7.4$. The threshold $\theta = 0.40$ is set below 0.5 so that a bilateral near-tie between the target and one neighbour (approximately 0.5/0.5) is scored correct, reflecting that the camera evidence cannot separate the two; and above $1/3$ so that an event fails whenever any other finger clearly dominates, and also in the rare three-way tie, where the visual evidence carries essentially no information about the acting finger.

Distances are measured to the axis-aligned bounding box of the pressed key’s pixel mask, which keeps the computation stable, with a separate per-event flag recording whether the fingertip also lies inside the exact mask. When several note-on events share a timestamp, candidate pairs are assigned greedily in order of decreasing probability so that one fingertip cannot be credited with two simultaneous notes, while each note’s distribution is still computed over all visible fingertips. This batch capability supports chordal material elsewhere in the platform, so the single-key protocol is an experimental restriction rather than a limit of the matcher. Events with no visible hand in the aligned frame remain unresolved rather than being forced to a label. The complete per-event distribution is stored in the results file, so the correctness criterion can be recomputed exactly under any alternative θ without reprocessing video; the live monitoring view and the offline scoring pipeline call one shared implementation of both the distribution and the threshold rule, so the two paths cannot drift apart.

Figure B.3 shows four of the events whose automatic label the reviewer overrode, the class behind the 4.50% correction rate of Section 3.2.1. Each is a near-tie the softmax resolved towards an adjacent fingertip, which is why the pre-screen was treated as evidence to be checked rather than as a label.

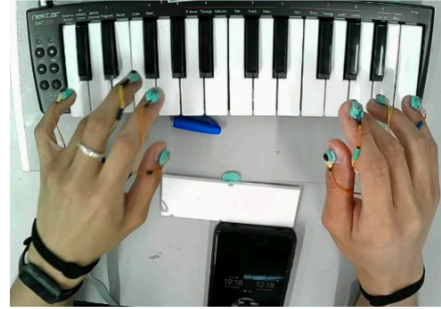
Four keypresses the automatic finger labelling got wrong and the reviewer corrected from the video. In each, the cued finger is hidden behind a neighbour, so both fingertips project onto the same key, the distance-based SoftMax splits between them, and the cued finger falls below the $\theta = 0.40$ threshold.

(a) cued R5 · detector read R2 · $p(R5) = 0.19$



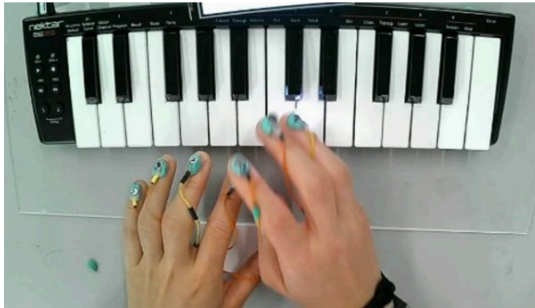
little finger curled in behind the hand

(b) cued L4 · detector read L3 · $p(L4) = 0.28$



ring finger behind the middle finger

(c) cued R5 · detector read R4 · $p(R5) = 0.31$



little finger behind the ring finger

(d) cued R3 · detector read R2 · $p(R3) = 0.38$



middle finger behind the index

Figure B.3. Four keypresses whose automatic finger label was corrected during review. Each panel header gives the cued finger, the finger the detector reported, and the probability the detector assigned to the cued finger.

B.8 Tele-training Measurement Protocol

Both teacher and student computers connected over Wi-Fi to Imperial College London's eduroam network. The relay server was located in London; no content delivery network (CDN) was used. The benchmark sent 1,000 cue-sized probes from teacher to student at fixed intervals after warm-up, counting acknowledgements and application-level loss; round-trip time used the teacher's monotonic clock. Eleven separate two-machine sessions measured student dispatch latency for all 244 unique timed cues and visual-haptic skew for the 160 with both timestamps. Both used the student's monotonic clock, unsynchronised with teacher and server clocks. The software timestamps recorded repaint scheduling and drive-command writes, not physical onsets.

Round trip was bimodal rather than heavy-tailed (Figure B.4). The link alternated between two states near 34 and 142 ms, so the 35.5% of probes above 100 ms reported in Chapter 3 record how long the slower state was occupied rather than isolated spikes. No probe was lost, and the round-trip median, 95th and 99th percentiles were 36.6, 150, and 258 ms.

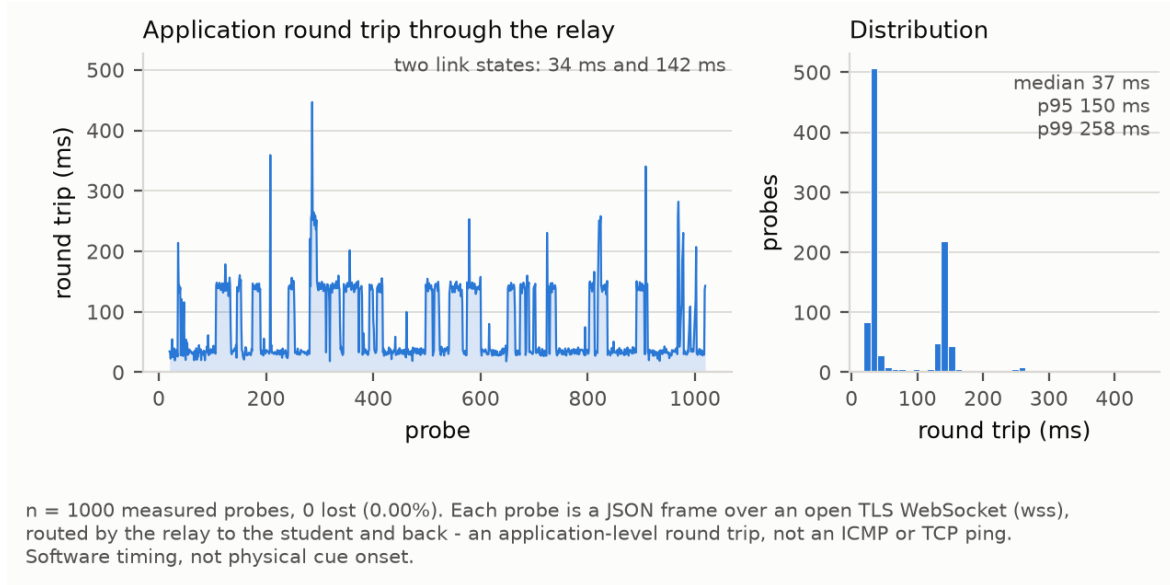


Figure B.4. Application round-trip time through the relay for the 1,000-probe benchmark of Section 2.4: (left) each probe in send order, (right) the resulting distribution. A probe is a cue-sized JSON frame sent over an already-open secure WebSocket, relayed to the student and back, so the quantity is an application-level round trip rather than a network-layer ping, and it reflects software timing rather than physical cue onset. No probe was lost. Median 36.6 ms, shown rounded to 37 ms in the right panel; 95th percentile 150 ms; 99th percentile 258 ms.

The typical note-on-to-cue budget in Chapter 3 summed half a tracking period, $RTT/2$, and median student dispatch ($77/2 + 36.6/2 + 1.1 \approx 58$ ms). Finger identity used the newest completed landmark pass (77 ms per two-hand 1280×720 pass with both clients on one host); $RTT/2$ assumes symmetric paths. Component p95 estimates summed to approximately 170 ms. This is not an end-to-end p95 because summing component percentiles does not recover the total's percentile; end-to-end latency was not measured.

Reaction time ran from cue readiness on the student's monotonic clock, excluding transport delay. Both demonstrations used the same automatic pipeline without frame review, one non-naive operator, unequal sequence lengths, and no counterbalancing.

Appendix C

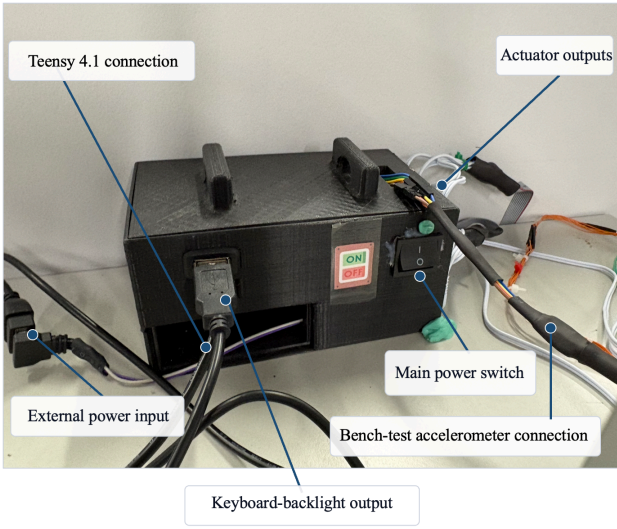
Reproducibility

The values below are read from the working system rather than from design notes, so a discrepancy between this table and the hardware is a fault in one of them and can be checked against the source. The host application and the embedded firmware are at <https://github.com/shengjiang/nail-haptic-piano-platform>.

Table C.1. Hardware and software manifest. Motor *ports* are logical channels on the driver boards and correspond one-to-one with the Teensy PWM pins of the same number.

Item	Specification
<i>Control</i>	
Microcontroller	Teensy 4.1
Firmware	<code>haptic-piano v2.10.0</code>
Host link	USB serial; host application in Python (PySide6)
<i>Haptic chain</i>	
Driver boards	$2 \times$ NhaptX v1.0, custom PCBs built around STSPIN240 dual brushed-DC drivers
Finger actuators	Ports 0–9, one per digit: L5→0, L4→1, L3→2, L2→3, L1→4, R1→5, R2→6, R3→7, R4→8, R5→9
Test channels	Port 10 ERM, port 11 LRA (wiring convention, overridable per session)
Drive settings	LRA 224 Hz at 25.1% PWM duty; ERM 1 kHz at 19.6% PWM duty; boot PWM frequency 224 Hz on every channel
<i>Visual cue</i>	
LED strips	$2 \times$ WS2812, GRB, 60 pixels each, data on pins 24 and 29 , driven by WS2812Serial (hardware UART + DMA)
<i>Measurement</i>	
Accelerometers	Up to $3 \times$ LIS3DH on one bit-banged SPI bus (mode 3, ≈ 2.5 MHz): SCK 33 , MOSI 34 , MISO 35 , CS 36/37/38
Keyboard	Nektar SE25, 25 keys, class-compliant USB MIDI
Camera	Unbranded USB webcam, driver-free (UVC), 1080p sensor; mounted overhead and operated at 1280×720 , 30 fps
<i>Power</i>	
3.3 V	Driver-board logic and LIS3DH, from the Teensy 3.3 V pin
5 V	WS2812 strips, from the Teensy Vin pin
9 V	Motor drive rail chopped by PWM, kept off the logic supply. Either a battery or a USB 5 V boost converter set to 9 V (1 A); the latter was used here. For duty $D = \text{amp}/255$, the time-averaged PWM voltage is $9D$ V (2.26 V at the selected LRA setting); the LRA’s vibration is governed mainly by the AC fundamental rather than this mean
Ground	All grounds tied common

(a) External interfaces



(b) Internal electronics

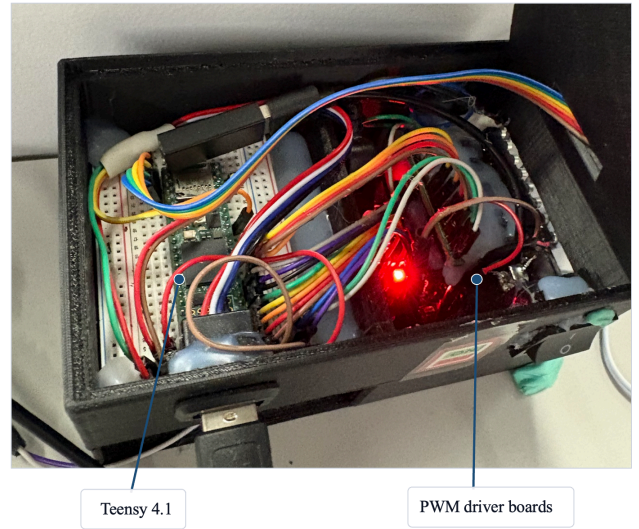


Figure C.1. Integrated 3D-printed control enclosure. (a) External interfaces: Teensy 4.1 connection, keyboard-backlight output, external power input and main switch, actuator outputs, and bench-test accelerometer connection. (b) Internal Teensy 4.1 and PWM driver boards. The enclosure protected the electronics while exposing the interfaces required during bench validation and user sessions; the accelerometer connection was used only for bench testing.

Two constraints are not obvious from the table. Teensy 4.1 inputs are not 5 V tolerant, so every signal line including the LIS3DH bus runs at 3.3 V. The WS2812 data pins are not free choices either: WS2812Serial requires a hardware UART transmit pin, and of the eligible pins the motor PWM outputs and the accelerometer bus already claim the lower ones, which is why the strips sit on 24 and 29.

The analysis export retains the field name `fa_main`; it corresponds to Complete-action Accuracy (CA), requiring both the key and finger to satisfy the scoring rule in Section 2.3.5.

C.1 Reaction-Time Dispersion

For each participant and condition in Figure 3.7, let r_j , s_j , and n_j denote the mean, sample SD, and count of valid complete-action reaction times for homologous digit j , pooling hands and trials. With all five digit cells available,

$$\begin{aligned} \bar{r} &= \frac{1}{5} \sum_{j=1}^5 r_j, & v &= \frac{1}{4} \sum_{j=1}^5 (r_j - \bar{r})^2, & \text{SEM}_j^2 &= \frac{s_j^2}{n_j}, \\ \text{SD}_{\text{corrected}} &= \sqrt{\max\left(v - \frac{1}{5} \sum_{j=1}^5 \text{SEM}_j^2, 0\right)}, & \text{CV} &= \frac{\sqrt{v}}{\bar{r}}. \end{aligned}$$

The correction subtracts estimated sampling-noise variance under independent errors in the digit means, truncating negative adjusted variances to zero. The plotted CV uses the uncorrected SD. Group summaries average participant values; the CV test compares paired C–B differences.

C.2 Computational Model and Held-out Prediction

Each resolved response was modelled as a softmax choice among the ten candidate fingers. For participant s , condition c , target key k_t , cued finger (h_t^*, d_t^*) , and candidate $f = (h_f, d_f)$,

$$U_{ts}(f) = \rho(h_f, k_t) + w_c[\pi(f) + \pi_s(f)] + \mu_1 I(f = f_{t-1}) + \mu_2 I(h_f = h_{t-1}) \\ + \kappa_{H,c} I(h_f = h_t^*) + \kappa_{D,c} I(d_f = d_t^*) - \gamma_c |d_f - d_t^*|, \\ P(F_t = f) = \frac{\exp U_{ts}(f)}{\sum_{g=1}^{10} \exp U_{ts}(g)}.$$

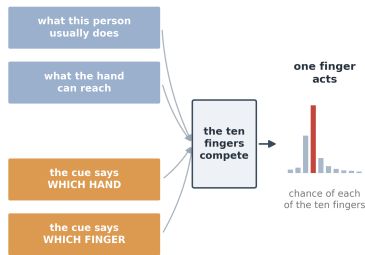
Condition A has no cue term. M0 removed cue evidence, M1 used a unitary cue, M2 separated hand and digit, M3 added the digit-distance gradient, and M4 freed evidence by cued digit. M3 was the primary interpretable model; M4 tested added flexibility. Penalised maximum-likelihood fitting used the target key, previous completed response, and training-participant data only. Population LOPO held out one participant per fold; Condition-A personalisation was evaluated separately.

Table C.2. Computational-model checks. Log loss is lower when prediction is better; all prediction rows are held out.

Check	Result	Reading
Primary fit	16,195 events; 250 parameters; converged	Descriptive fit only; not evidence of generalisation.
Cue contrasts	Hand: B 5.165, C 8.820; digit sharpness: B 3.584, C 3.744	The C hand estimate is a lower bound: only two C actions used the wrong hand. No ratio is interpreted as an effect size.
LOPO choice prediction	M3 log loss 0.3053; M4 0.2963; multinomial 0.2963; M3–multinomial paired $p = .156$	M3 improved reduced variants, but not generic predictive accuracy.
Homologous-error ranking	M3 captured 35/53, 47/53, and 51/53 in the top 10%, 20%, and 30%; at 20%, M4 captured 51/53 and multinomial 50/53	Risk generalised to unseen participants, but the enrichment was not unique to M3.
Within-hand ranking	38/359 in M3's top 20%	No predictive lift; this broad detected-mismatch class is not the genuine-substitution class used in the confusion analysis.
Event-level RT	Population M3 R^2 : A 0.009, B 0.026, C 0.054	Event prediction was weak; the measured 247 ms B–C difference was not derived by the model.
Condition-A personalisation	Δ log loss = +0.0024, 95% CI $[-0.0039, 0.0087]$, $p = .432$	No improvement in cross-participant prediction.

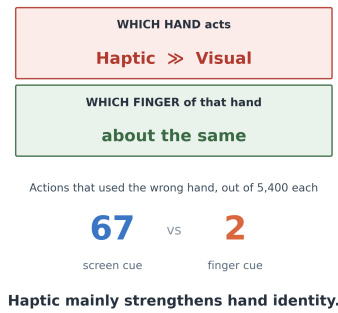
Figure C.2 links the model structure to the cue-evidence estimates and held-out ranking summarised in Table C.2.

A. How the model chooses a finger



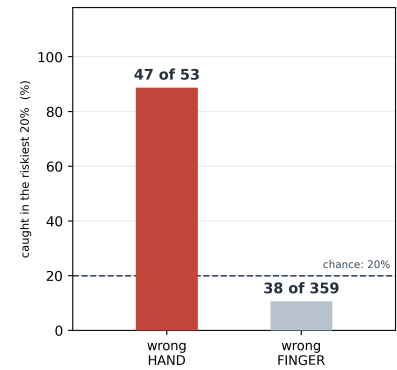
Nothing in the model mentions "wrong hand". It is told only whether a candidate matches the cued hand and whether it matches the cued finger.

B. What the haptic cue strengthens



Direction and raw counts, not a ratio: under the finger cue the wrong hand was used twice in 5,400 events, which fixes that channel only from below.

C. Predicting mistakes in advance



Wrong-hand mistakes are predictable in advance — and simpler models rank them about as well.

Held out one participant at a time; every input is known before the response.
Within-hand counts every detected mismatch, a wider class than the report's genuine substitutions.

Figure C.2. Computational model and held-out prediction. (A) Ten-finger competition. (B) Estimated cue evidence for hand identity and within-hand digit. (C) Population leave-one-participant-out ranking. The grey bar includes detected mismatches rather than genuine substitutions; generic multinomial prediction ranked wrong-hand risk similarly.

Appendix D

Secondary Behavioural and Deployment Results

This appendix retains five supporting visualisations that provide additional context for Chapter 3 without altering its primary inferences, followed by the unmatched single-operator sessions used only to verify the complete remote path.

D.1 Additional Behavioural Visualisations

Figure D.1 shows the participant trajectories behind the condition-level accuracy summaries.

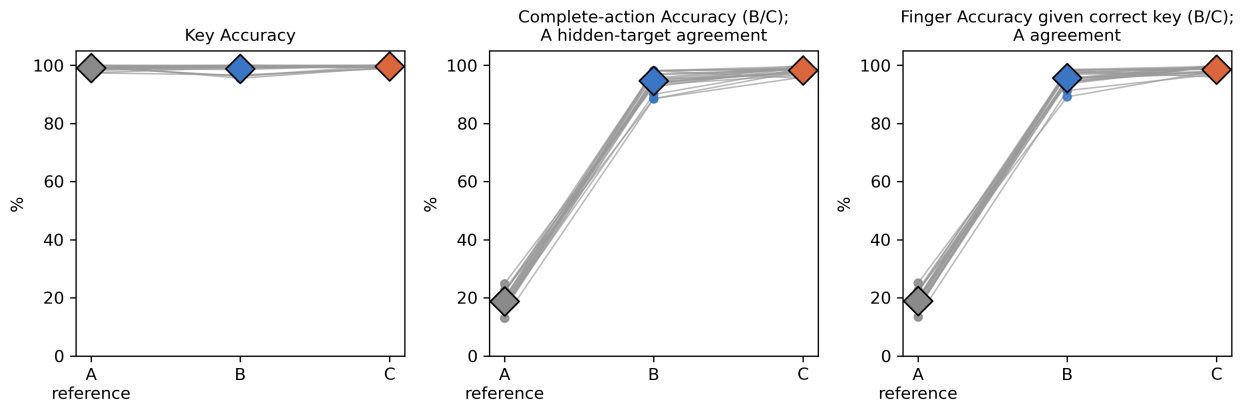


Figure D.1. Accuracy outcomes by condition. Thin lines are participants; diamonds and bars are group means and 95% participant-level bootstrap CIs ($n = 20$). In A, finger measures are agreement with the hidden generated fingering, not accuracy to a supplied cue.

Figure D.2 shows the per-digit Complete-action Accuracy cells summarised in Chapter 3. The outcome is ceiling-bounded and was not entered into the ANOVA; the binomial mixed model fitted to these same cells is Figure 3.8 in the main text.

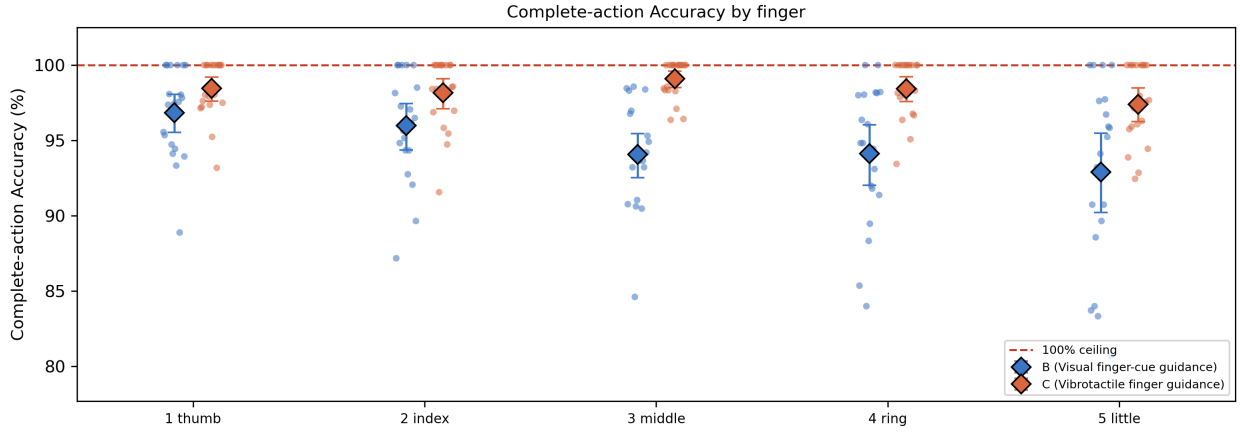


Figure D.2. Complete-action Accuracy by homologous finger. Points are participant cell means; diamonds and bars are group means and 95% participant-level bootstrap CIs ($n = 20$). This ceiling-bounded outcome is descriptive and was not entered into the ANOVA; the model fitted to the same cells instead is Figure 3.8.

Figure D.3 shows participant-level rates for correct-key events assigned to a different finger. In B and C this is a cueing error; in A it is disagreement with the hidden generated reference.

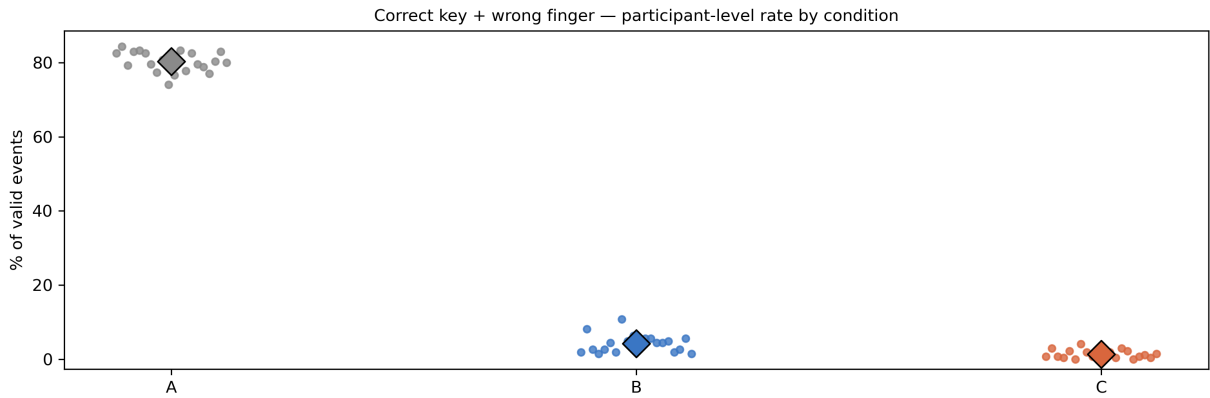


Figure D.3. Participant-level proportion of valid events with the correct key but a different finger. In B and C, “different” means different from the supplied cue; in A, it means different from the hidden generated reference and is not an error against an instructed finger. Points are participants; diamonds and bars are group means and 95% participant-level bootstrap CIs ($n = 20$).

Figure D.4 complements the controlled Condition $\times$ Repetition analysis by aligning, within each participant and difficulty, the nine trials by actual session order. From the first to ninth occurrence, observed group-mean reaction time fell from 733 to 581 ms for α , 901 to 613 ms for β , and 989 to 743 ms for γ , while key accuracy stayed between 98.0 and 100.0%. Because the randomised condition mix differs across occurrences, these raw trajectories are descriptive rather than estimates of a practice effect; condition-adjusted values remain available in the tidy export.

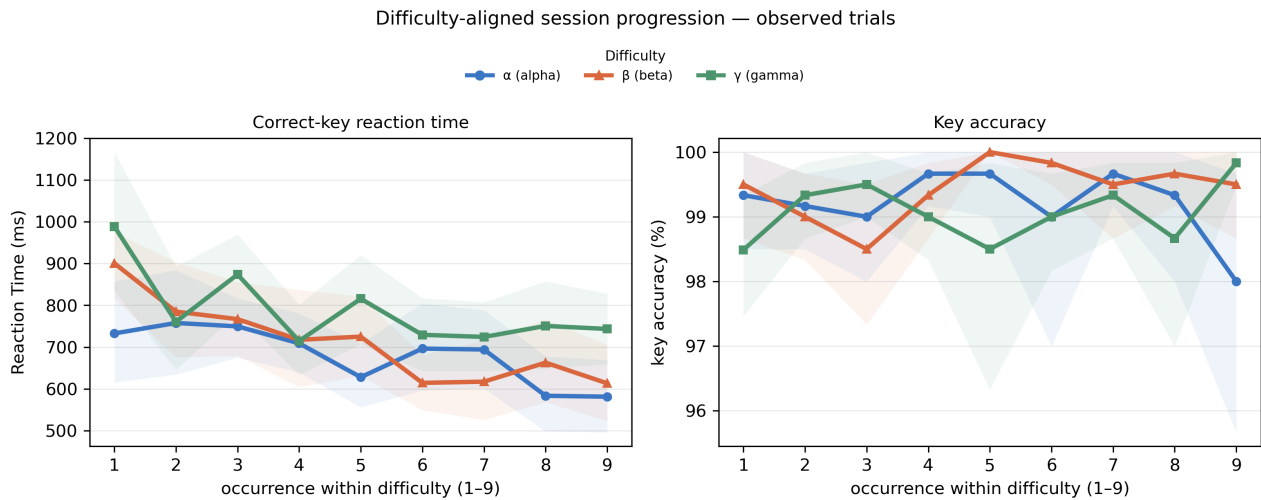


Figure D.4. Difficulty-aligned within-session progression across all A/B/C trials. Within each participant and difficulty, trials are sorted by actual session order and relabelled occurrence 1–9. Marked lines are group means and shaded bands are 95% participant-level bootstrap CIs ($n = 20$). Panels show observed correct-key reaction time and key accuracy. Because condition composition varies across occurrences, the trajectories are descriptive; condition-adjusted values are exported but not plotted.

Figure D.5 shows the participant-level hand preference underlying the Condition A summary in Chapter 3.

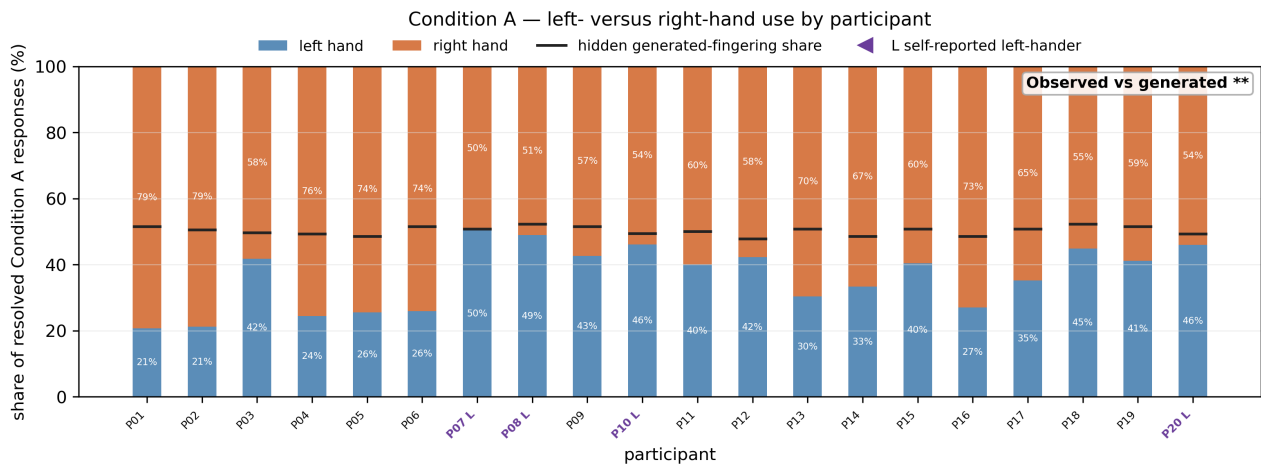


Figure D.5. Left- versus right-hand share of resolved Condition A presses by participant. Horizontal rules mark the hidden generated-fingering share; purple “L” labels mark the four self-reported left-handers. The overall observed-versus-generated annotation uses a two-sided paired t -test ($*p < 0.05$, $**p < 0.01$); comparisons by handedness remain descriptive ($n = 20$).

D.2 Tele-training Demonstration Sessions

Two single-operator sessions exercised the remote path. Visual guidance produced 13/16 correct actions (mean 951 ms), and vibrotactile guidance 44/47 (500 ms). With one non-naive operator, no counterbalancing, and no frame-by-frame review, these unmatched sessions verify delivery but not behavioural benefit.

Appendix E

Originality Statement

This dissertation is the author's own original work. The author independently formulated the research problem, designed and implemented the system, developed the experimental methodology, conducted the study, analysed the data, interpreted the findings, and wrote the conclusions. The intellectual contribution, technical decisions, experimental results, and arguments presented in this dissertation are therefore the author's own.

Artificial intelligence tools were used only as a limited editorial aid during preparation of the manuscript, principally for language polishing, grammar correction, and refinement of phrasing. They did not determine the research questions, methodology, implementation, data analysis, interpretation, or conclusions. All suggested wording was critically reviewed and revised by the author, who retains full responsibility for the accuracy, originality, and final content of the dissertation.